Why Did the Ranker Pick the Wrong Title? Explaining Recommendation Failures to Decide What to Fix

Counterfactual (TemGX, ICLR 2026) and causal (CATGX, KDD 2026) explanations over members' rating history

Mingjian Lu

PhD candidate, Case Western Reserve University

[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026.

One failed search, from the symptom to a fix worth testing

A search goes wrong
"a fantasy film for grown-ups" → Aladdin is #1

→

Where does it fail?
Empirical study on a ranker I built

→

Which past events?
the few past ratings the model relied on (TemGX)

→

Which patterns?
recurring patterns that matter beyond timing (CATGX)

→

Which fix to test?
combine the evidence; check a fix before an A/B test

→

Demo and next steps
Live demo; where this can go

The investigation in five steps: it stays on the right of every slide as navigation

1 Failure | 2 Events | 3 Patterns | 4 Together | 5 Fix

Search = a member's request to the ranker, the model that orders titles. **Failed search** = the #1 title does not fit the request.
One running case throughout: member 304 on public MovieLens data.

Methods: [1] Lu et al., ICLR 2026; [2] Lu et al., KDD 2026. Data: MovieLens [5].

Mingjian Lu

PhD student, Case Western Reserve University · Advisor: Professor Yinghui Wu



What evidence does a model rely on?

My research: discover which past evidence a model's prediction depends on, summarize it into cause-aware patterns, and choose the method that fits the context; then turn it into a fix worth testing.

Q1 Discover: which past events drive one prediction?

TemGX · Training-free Counterfactual Explanation for Temporal Graph Model Inference · ICLR 2026 [1]

Q2 Summarize: which recurring patterns really matter?

CATGX · Causal-Aware Temporal Graph Explanation via Scalable Motif Sampling and Adjustment · KDD 2026 [2]

Q3 Contextualize: does the best method depend on context?

Context Determines Optimal Architecture in Materials Segmentation · ICLR 2026 AI4Mat workshop [13]

Selected publications: [1,2,13]. Photograph supplied by Mingjian Lu.

A member asks for a genre, with an implied exclusion

“A fantasy film for grown-ups”

The ranker scores every title by adding three signals:

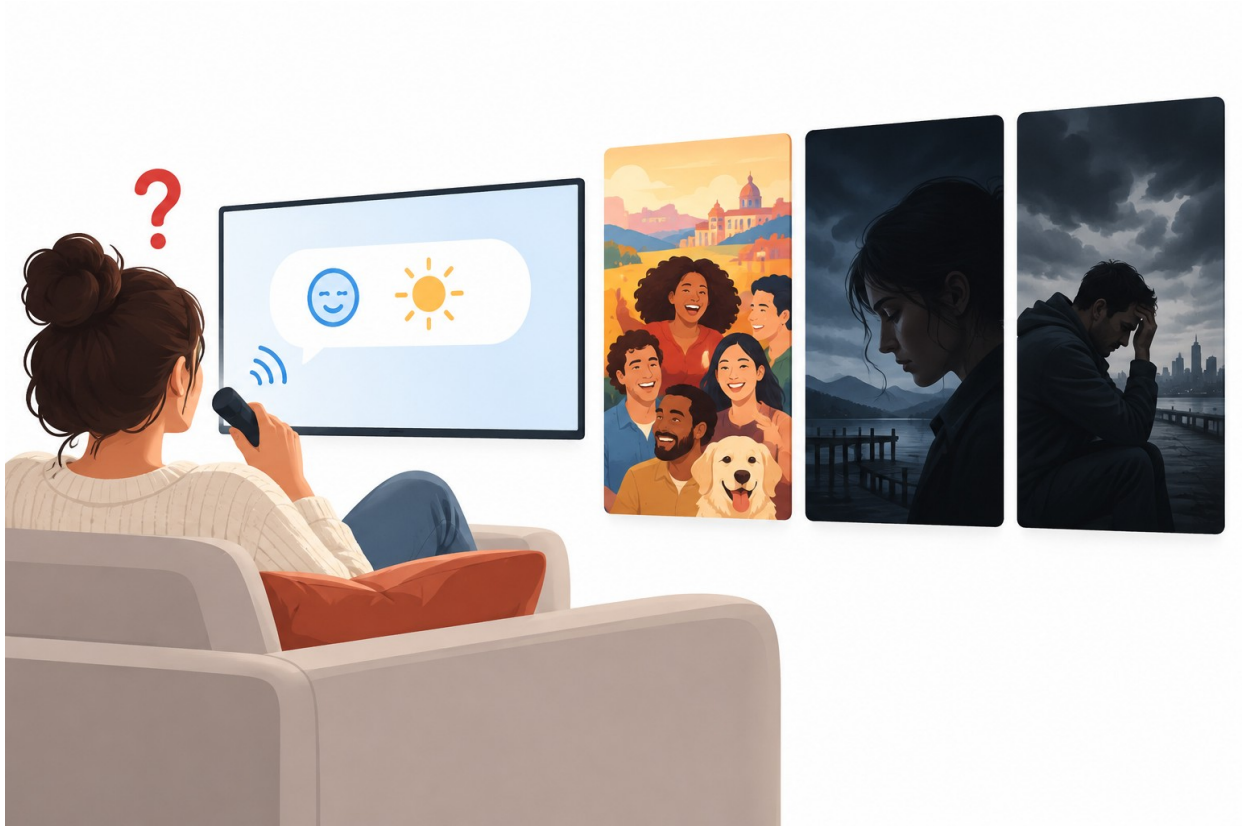
Text match does the title fit the words of the request?

Popularity how many members watch it

History what this member and similar members rated recently

Text match already prefers Harry Potter; **popularity** and **history** put Aladdin ahead.

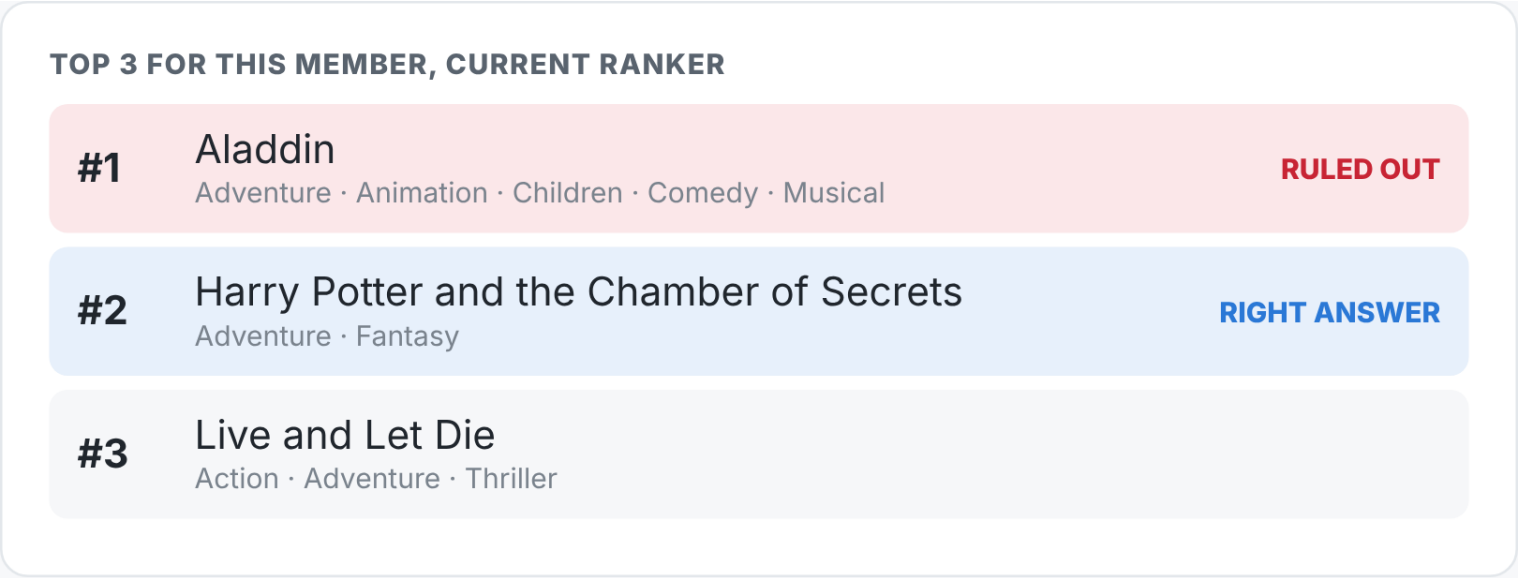
None of them reads that “for grown-ups” rules out a family cartoon, so Aladdin comes first



[3] Gomez-Uribe & Hunt, ACM TMIS, 2015. Ranker and signals: the Demo on MovieLens [5], member 304.

The top result contradicts the member’s request

Member 304: “A fantasy film for grown-ups”



#1

is a family cartoon: “for grown-ups” rules it out. Right answer: #2

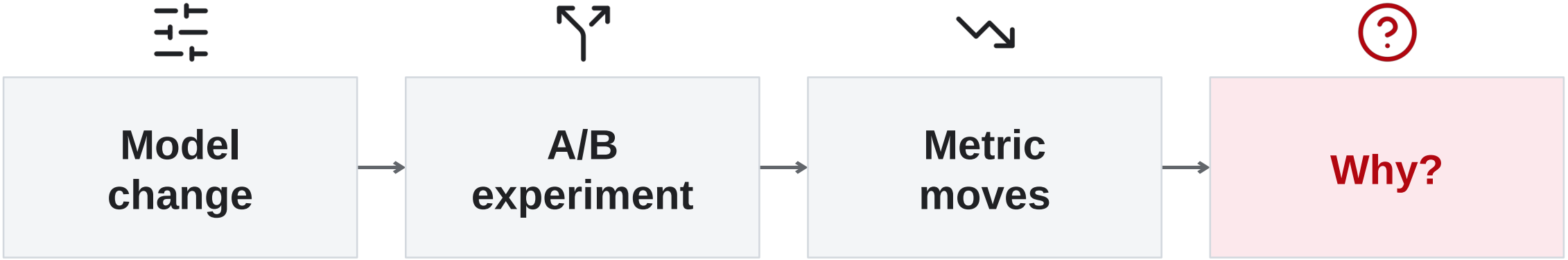
Right answer: a title that fits the request (here a live-action fantasy).

Why does Aladdin win, and what should the team change?

Dataset: MovieLens [5]. Demo, member 304.

An A/B test shows that a metric moved, not why Aladdin won

Model = the ranker: request + rating history in, ranked titles out. An A/B test compares two rankers on a metric, e.g. how often members play a top result.

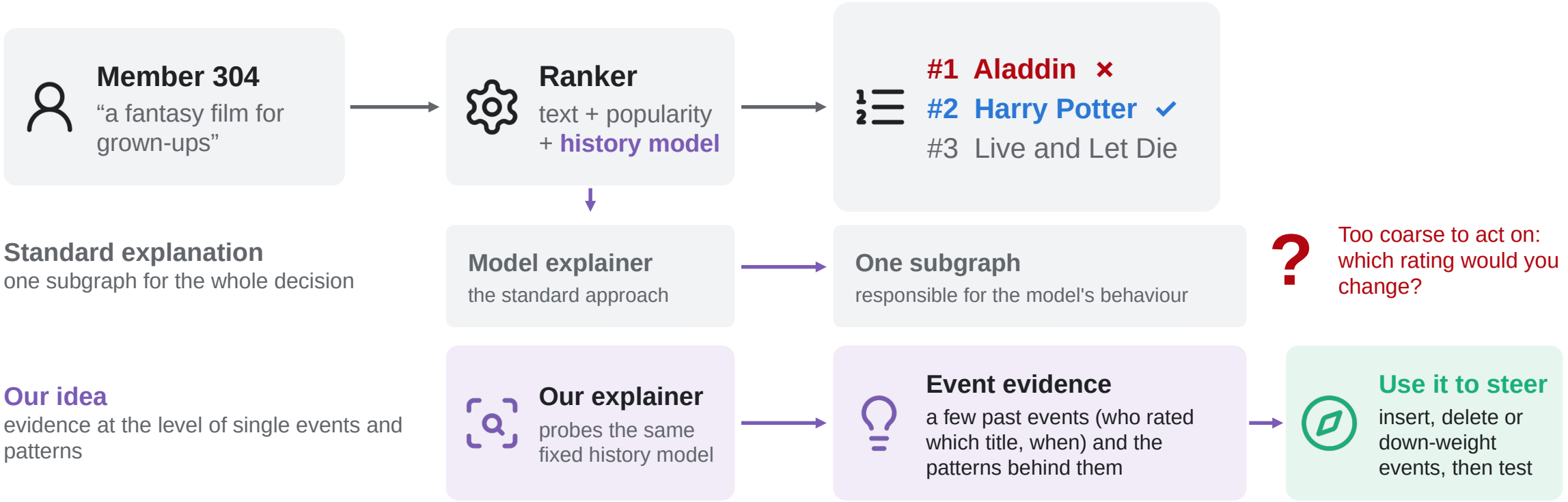


Which requests fail? Which past events made Aladdin win?

An A/B test reports one aggregate number over all members; it cannot say which of member 304's past ratings pushed Aladdin up.

[3] Gomez-Uribe & Hunt, 2015. [4] Kohavi et al., 2009.

Idea: explain the wrong ranking with the evidence it depends on



Discriminative: removing the event evidence must change the fixed history model's preference between the wrong #1 and the right answer.

Illustration of the approach. Ranker and data: Demo on MovieLens [5].

A five-step investigation, from a failed search to a fix worth testing

1 Find the failure

Empirical study: which requests fail, how often, and which ranker signal caused it

In: logged searches
Out: failing requests

2 Which events?

TemGX: a compact set of past ratings whose removal changes the history model's preference

In: one failed search
Out: the key events

3 Which patterns?

CATGX: recurring patterns that still matter when compared only with equally recent, equally active ones

In: rating history
Out: patterns that matter

4 Put together

Combine the events, the patterns and how much each ranker signal contributed

In: steps 1–3
Out: a diagnosis

5 Decide what to test

Reuse fixes that worked on similar, already-diagnosed failures; check on new requests

In: diagnosis
Out: a fix to A/B test

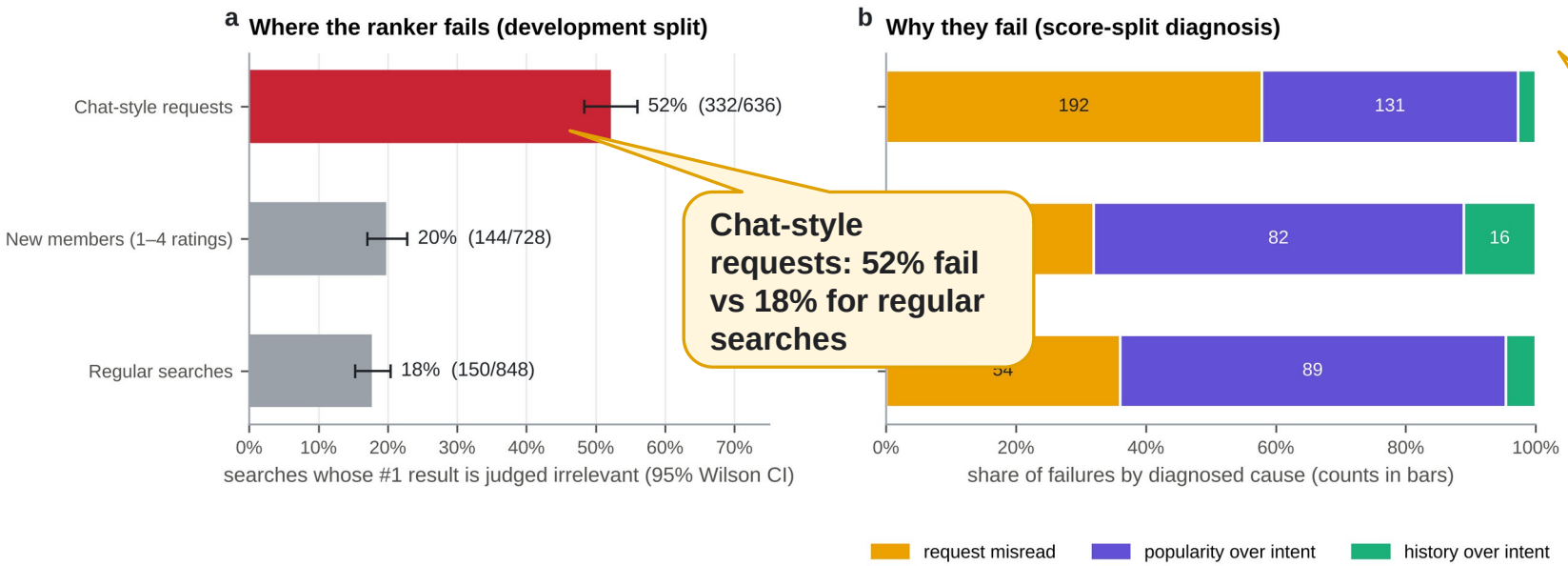
Each step can run as an agent: an event agent (step 2), a causal-analysis agent (3) and an explanation agent (4–5).

I built a Demo that runs all five steps on a failed search; I'll show it after the methods.

Methods: [1,2]. Demo on MovieLens [5].

Empirical study: chat-style requests fail three times as often

Setup: on public MovieLens I built a ranker (text match + popularity + a model of past ratings, the “history model”) and ran 2,212 searches. Failure: the #1 result does not fit the request. Cause: the signal that gave the wrong title its lead.



Chat-style requests: 52% fail vs 18% for regular searches

Most failures: the request is misread or popularity overrides intent

So what: we know where it fails, but not which past events made the wrong title win

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Experiment: Demo, development split. Chat n = 636; regular n = 848; Wilson 95% intervals.

An important-looking event can be the wrong evidence



Member 304 · “a fantasy film for grown-ups”

The model prefers **Aladdin** over **Harry Potter: 73%**

The 3 ratings with the highest attention weight:

- user 533 rated Aladdin 4.5★ · Apr 4, 2016
- user 138 rated Harry Potter 3★ · Jan 27, 2016
- member 304 rated The Usual Suspects 5★ · Apr 8, 2016

Remove all three → **74%**: the model still prefers Aladdin

Event = one rating: (user, title, stars, time). High attention weight is not the same as the evidence the decision depends on.

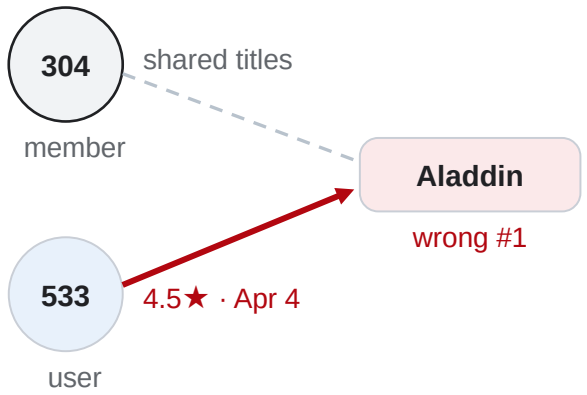
What evidence would explain why Aladdin wins?

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[9] Jain & Wallace, 2019. [10] Wiegreffe & Pinter, 2019. Demo, member 304, three-rating removals.

Three questions, each answered by its own tool

Which events?

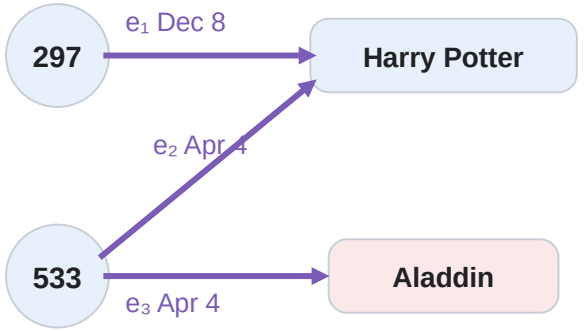


Single past events, e.g. one rating that made Aladdin look current

TemGX

finds a compact set of past events whose removal changes the history model's preference

Which pattern?



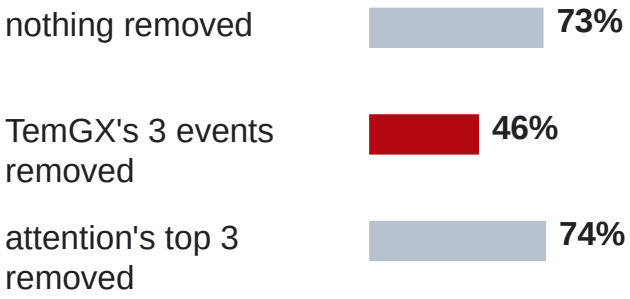
A recurring structure with an order ($e_1 < e_2 \leq e_3$): a Harry Potter fan's taste leads another member to Aladdin. Is it the structure, or just recent, active users?

CATGX

tests whether a pattern still matters after matching on timing and activity

Does it change the history model?

History model prefers Aladdin:



Below 50%, the history model ranks Harry Potter higher

Check

remove the evidence, re-inference, compare with attention's top picks

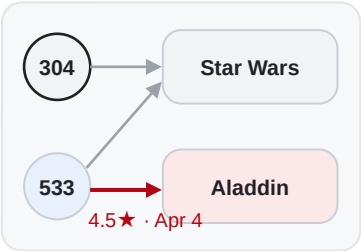
1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026. Values: Demo on MovieLens [5], member 304.

Problem: which past events drive the history model?

History graph G_t

Members and titles are nodes. Each edge is an event e = (user, title, stars, time τ), with τ before the search time t.



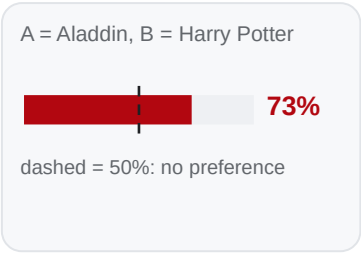
Ranker

score(i) = weighted sum of standardized text(q, i), popularity(i) and h(u, i; G_t); h is the history model, a 2-layer temporal graph attention network (TGAT).



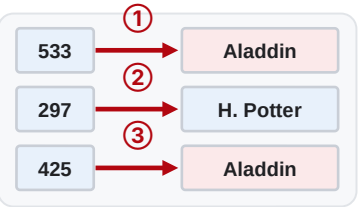
Target y

y = P(A above B) = σ((h(u, A) − h(u, B)) / T); A = wrong #1, B = right answer, σ = logistic, T = 0.65 (fitted on validation pairs).



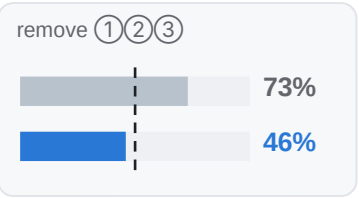
Explanation

A set of events $E_s \subseteq G_t$ with $|E_s| \le k$.



Counterfactual

Keep the trained model fixed; remove E_s; re-inference. If $y(G_t - E_s) < 50\%$, the model no longer prefers A.



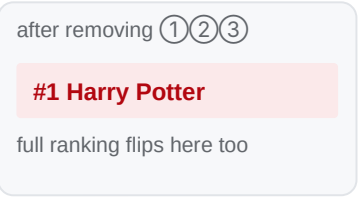
Control

Compare with removing the k events that attention weights highest.



Scope

y is about the history model; the full ranking also adds text match and popularity.



1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

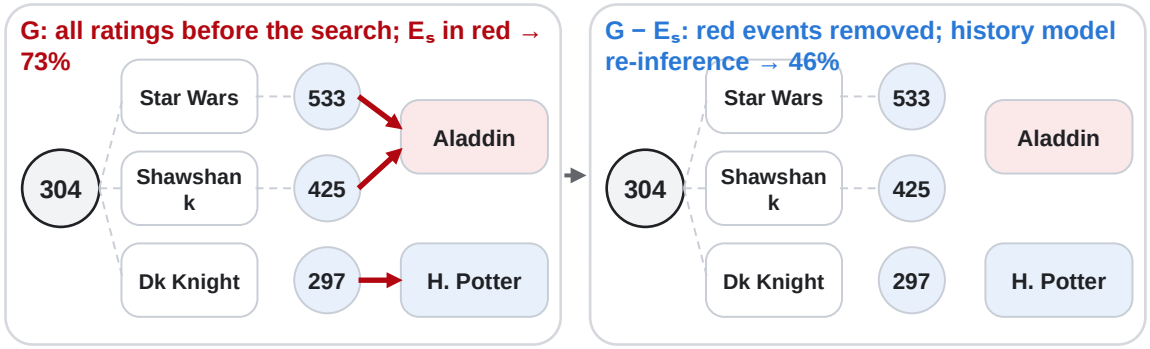
Formulation follows [1]. Mini figures: running values from the Demo on MovieLens [5], member 304.

Counterfactual explanation under a fixed model

Question

Would the same trained history model prefer differently if a small part of the history were absent?

Counterfactual explanation: a compact set of past events whose removal changes the fixed history model's preference.



Strict decision change

$$\hat{y}\theta(G - E_s) \neq \hat{y}\theta(G)$$

ŷ: does the history model prefer Aladdin over Harry Potter (y > 50%)? θ: its trained weights, fixed.
E_s: the removed events (red, left).

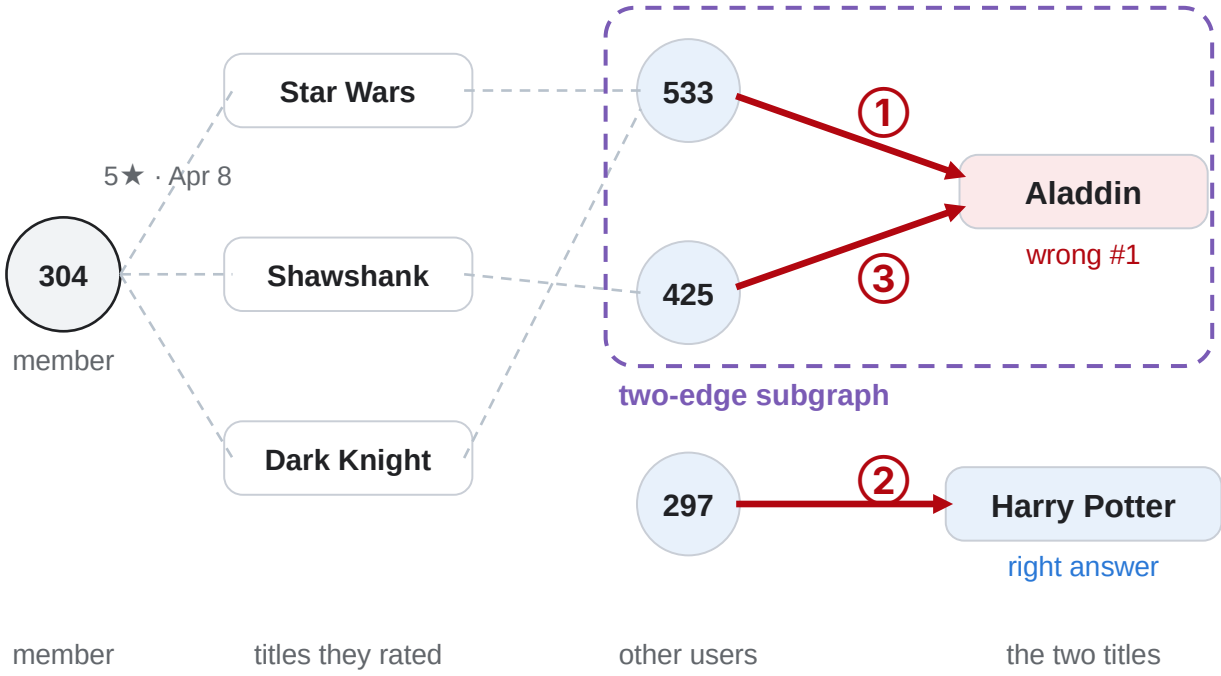
73% → 46%

Member 304: ŷ(G) = Aladdin at 73%; ŷ(G − E_s) no longer prefers Aladdin (46%)

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026. [6] Wachter et al., 2018. Example: Demo on MovieLens [5], member 304, k = 3.

TemGX finds a counterfactual explanation in terms of events



Selected events (k = 3)

① **User 533 rated Aladdin 4.5★, Apr 4, 2016 (4 days before)**

② **User 297 rated Harry Potter 3★, Dec 8, 2015**

③ **User 425 rated Aladdin 4.5★, Dec 27, 2015**

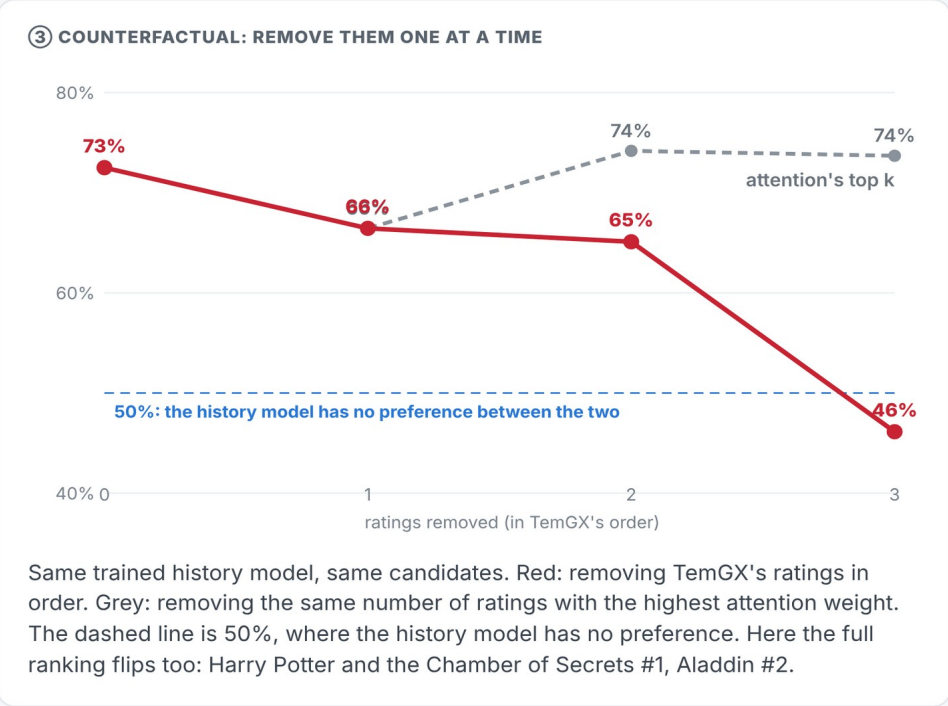
① + ③ act together. Two fans rating Aladdin months apart form a two-edge subgraph. Removed together: 73% → 48%; each alone: 66% and 72%.

Grey dashed: these users also rated titles member 304 rated, which ties them to the member.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026, §§3–4. Demo on MovieLens [5], member 304, k = 3. Removal values: same trained model, re-inference with the events deleted.

Case study: removing 3 events flips the history model



Same trained history model, same candidates. Red: removing TemGX's ratings in order. Grey: removing the same number of ratings with the highest attention weight. The dashed line is 50%, where the history model has no preference. Here the full ranking flips too: Harry Potter and the Chamber of Secrets #1, Aladdin #2.

y-axis: the history model's probability of ranking Aladdin above Harry Potter (as on the problem slide); 50% = no preference. **x-axis:** ratings removed one at a time; grey dashed: the same number of top-attention ratings.

Full ranking, same request (top 3, total score)

Before removal	After removing TemGX's 3 events
#1 Aladdin	**#1 Harry Potter**
#2 Harry Potter	**#2 Aladdin**
#3 Live and Let Die	#3 Live and Let Die
History part: **favours Aladdin**	History part: **favours Harry Potter**

Text match already favours Harry Potter; with the history part flipped, so does the total.

Removing them flips the history model: 73% → 46% (attention's top 3: 74%). Aladdin's lead was small, so the full ranking flips too.

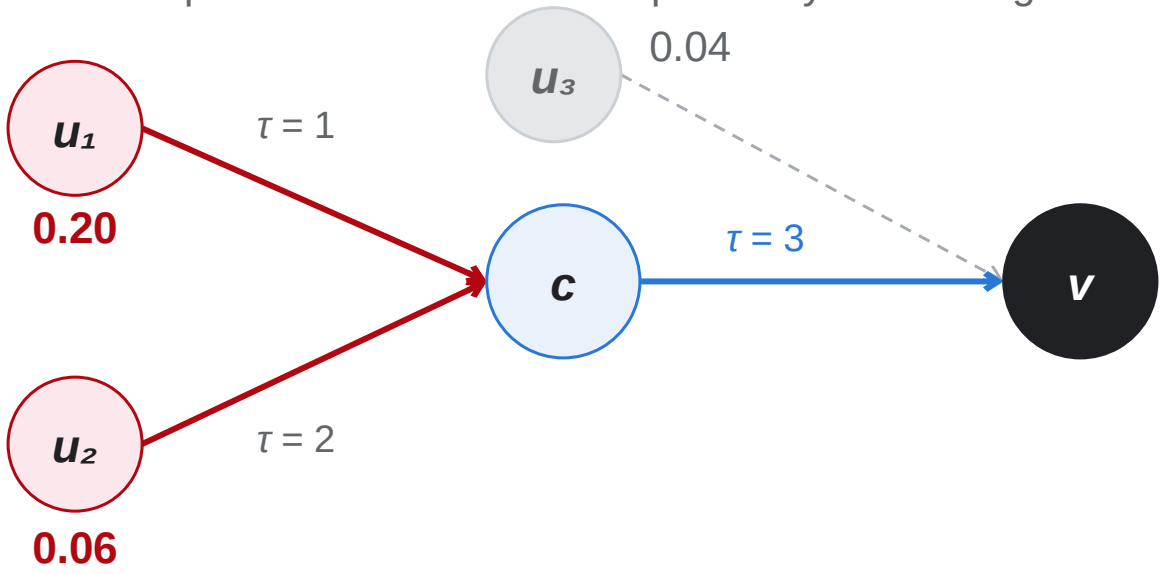
1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026. Demo on MovieLens [5]: member 304, k = 3; grey dashed = the same number of ratings with the highest attention weight.

TemGX picks explanatory nodes under a budget k

maximize $\Phi(V_s, v)$ **subject to $|V_s| \le k$**

Φ = total temporal influence k = explanatory-node budget



Toy (k = 2): picks u_1, u_2; connector c keeps a time-valid route to v; u_3 not picked.

Same optimization on member 304 (k = 3)

Greedy by Φ, each pick checked by re-inference:

- user 533 highest Φ ✓ kept
- user 138 2nd highest × no drop, rejected
- user 297 ✓ user 425 ✓

V_s = {533, 297, 425}: the 3 event-evidence ratings

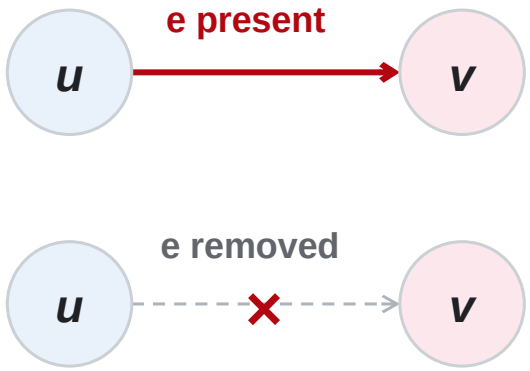
1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026, §3. Toy graph. k counts explanatory nodes; connectors preserve temporal paths.

Influence: remove one rating, then combine paths

$$p(e) = |\Pr(G,v) - \Pr(G \setminus \{e\},v)|$$

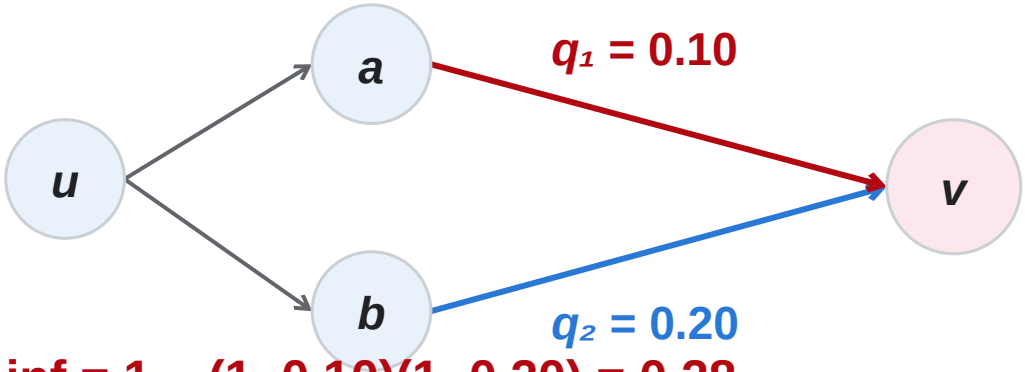
1 Remove one edge



$$p(e) = |0.80 - 0.70| = 0.10$$

2 Combine incoming paths

q = edge influence × upstream influence



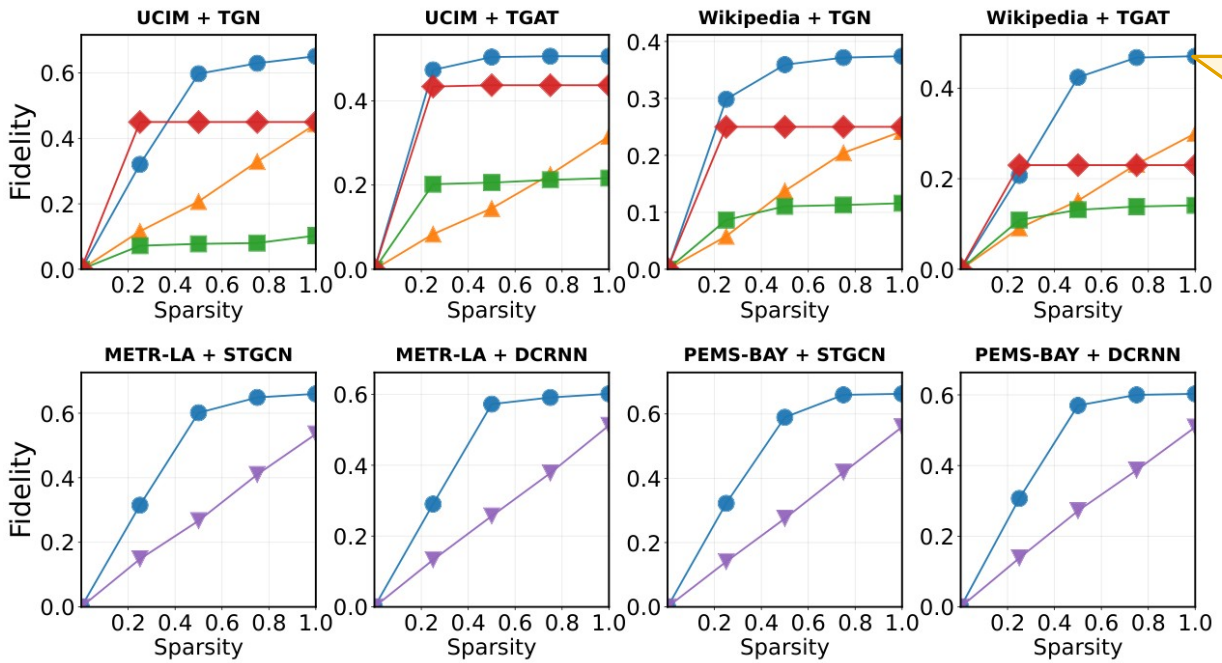
$$\text{inf} = 1 - (1-0.10)(1-0.20) = 0.28$$

The same three steps on member 304 (target: the history model's preference for Aladdin, 73%):
① remove user 533's Aladdin rating and re-inference; ② combine its time-valid paths; ③ weight by connectedness and recency → **Φ(533) is the highest of 36 candidates.**

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026, §3. Edge-removal sensitivity and ICM-style influence propagation.

TemGX: fidelity across explanation budgets



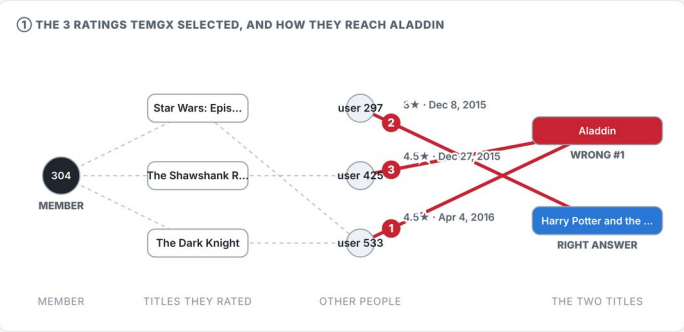
TemGX (blue) reaches the highest fidelity at most budgets, across 8 dataset–model pairs

Data: Wikipedia edits, UCIM messages, METR-LA / PEMS-BAY traffic sensors.
Protocol: matched backbones and explanation budgets, 5 independent seeds.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026, §5 and Fig. 4. Original fidelity panels; matched models and explanation budgets.

From selected ratings to a written diagnosis, with RAG



TemGX, k = 3: ① ② ③ are the red edges

Retrieved evidence, each rating checked by removal

①	User 533 rated Aladdin 4.5★ on Apr 4, 2016, **4 days before the search**; also rated **Star Wars** and **The Dark Knight**, which member 304 rated	remove only this: **−6.1 points**
②	User 297 rated Harry Potter **3★** on Dec 8, 2015; it changes how Harry Potter is represented	remove only this: **−1.2 points**
③	User 425 rated Aladdin 4.5★ on Dec 27, 2015; also rated **The Shawshank Redemption**, which member 304 rated	remove only this: **−0.6 points**

All three removed: 73% → 46%; attention's top 3: 74%.

RAG over verified diagnoses

1 Retrieve
TemGX + CATGX pick the event evidence; HNSW finds the 3 most similar of 341 verified diagnoses

→

2 Augment
One evidence sheet: the ratings, the pattern, the measured effects and the fixes that worked before

→

3 Generate
An LLM writes the diagnosis for the team, grounded in event evidence ①②③ and the 3 past cases

Generated diagnosis, member 304: *"Taking away Aladdin's ratings dropped how sure the model was to 46%, so those ratings drove the pick. … Try "understand what the request rules out", which worked for 18 of 20 of the most similar past failures."*

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[1] Lu et al., ICLR 2026. RAG [14]; GraphRAG [15]. Demo on MovieLens [5], member 304, k = 3. Retrieval: HNSW over 341 verified diagnoses.

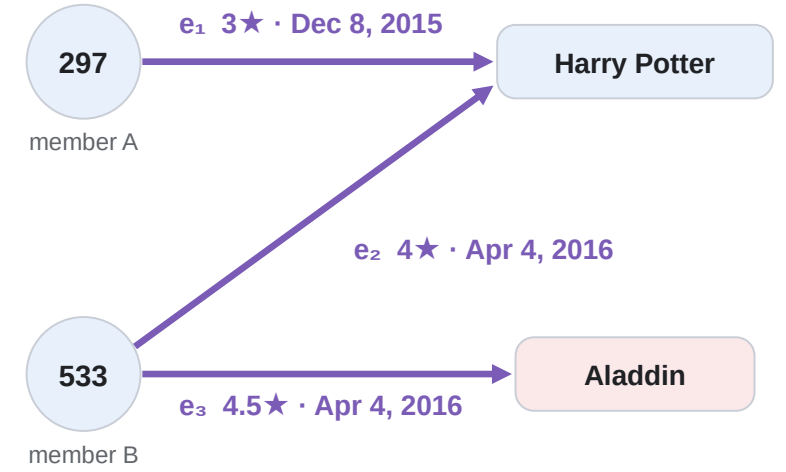
CATGX: patterns beyond timing and activity

A pattern (temporal motif): a small shape of ratings in a fixed order.

Pattern

Shared taste leads to the title: member B rates the same title as an earlier member A, then rates another title the same day.

Occurrence in member 304's neighbourhood



Order: $e_1 < e_2 \le e_3$ (three edges, two members, two titles)

Does the pattern matter, or are the participants simply recent and active?

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

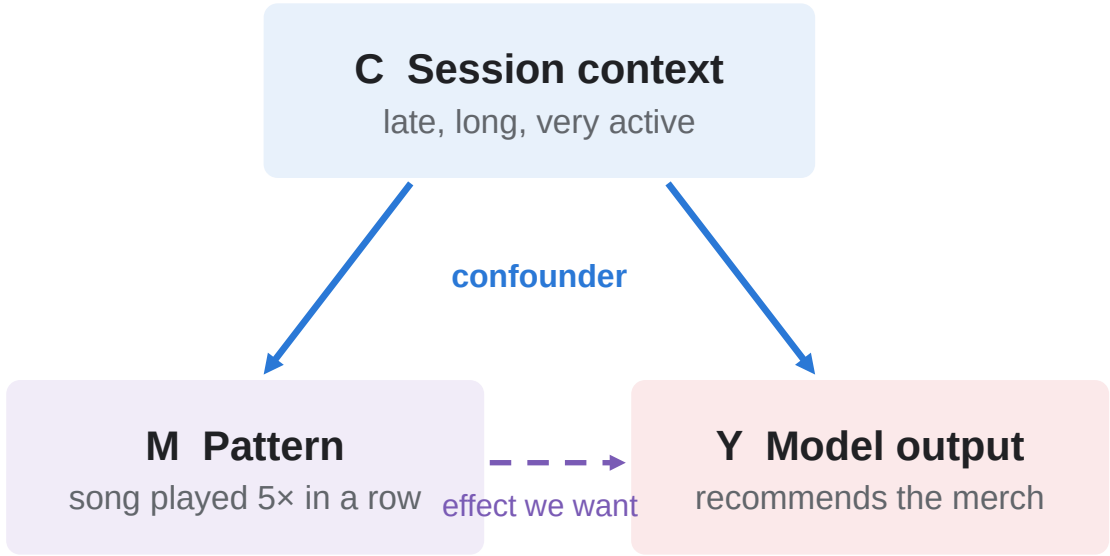
[2] Lu et al., KDD 2026, §§2–3. [8] Paranjape et al., WSDM 2017. Case occurrence: Demo on MovieLens [5].

A pattern can look important partly because of its session

A small story

21:02	Rates a comic-based film 5★
21:30–22:10	Plays its theme song 5 times in a row
22:12	Same page shows a merchandise ad
22:15	Model recommends the merch title

The member is not interested in the merch: the recommendation is wrong.

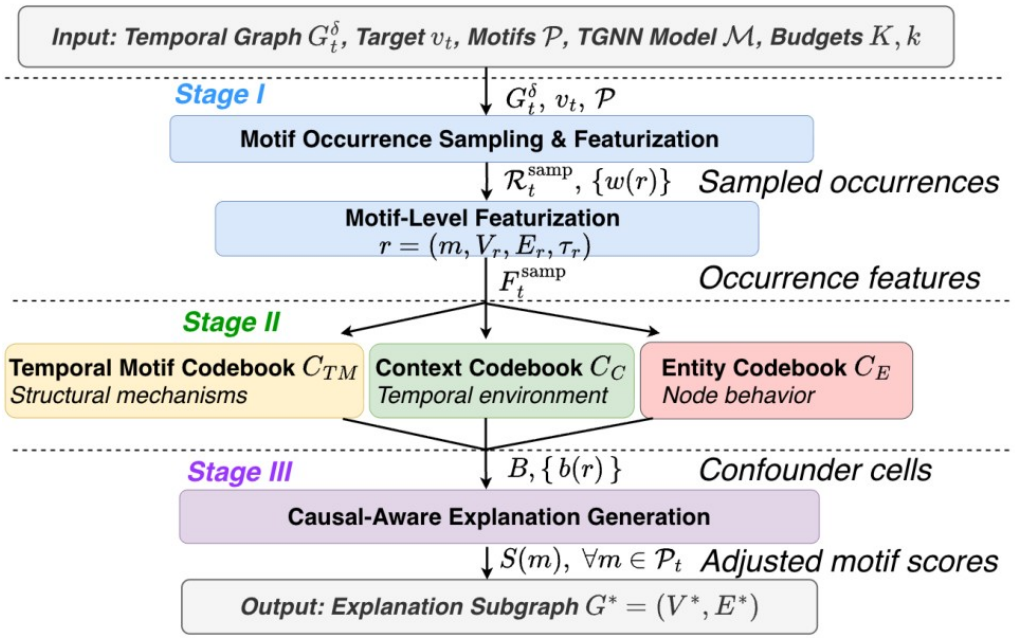


Backdoor path $M \leftarrow C \rightarrow Y$

Block it by comparing M only within the same context C: **backdoor adjustment**

[2] Lu et al., KDD 2026, §3 and §6. [7] Pearl, Statistics Surveys, 2009.

CATGX compares a pattern only within matched context cells



CATGX: sample pattern occurrences → context cells → adjusted score

Backdoor adjustment: **P(Y | do(M = m)) = Σc P(Y | M = m, C = c) P(C = c)**

$$S(m) = \sum_b \omega(b)\,[\,\mu(m,b) - \mu_{base}(b)\,]$$

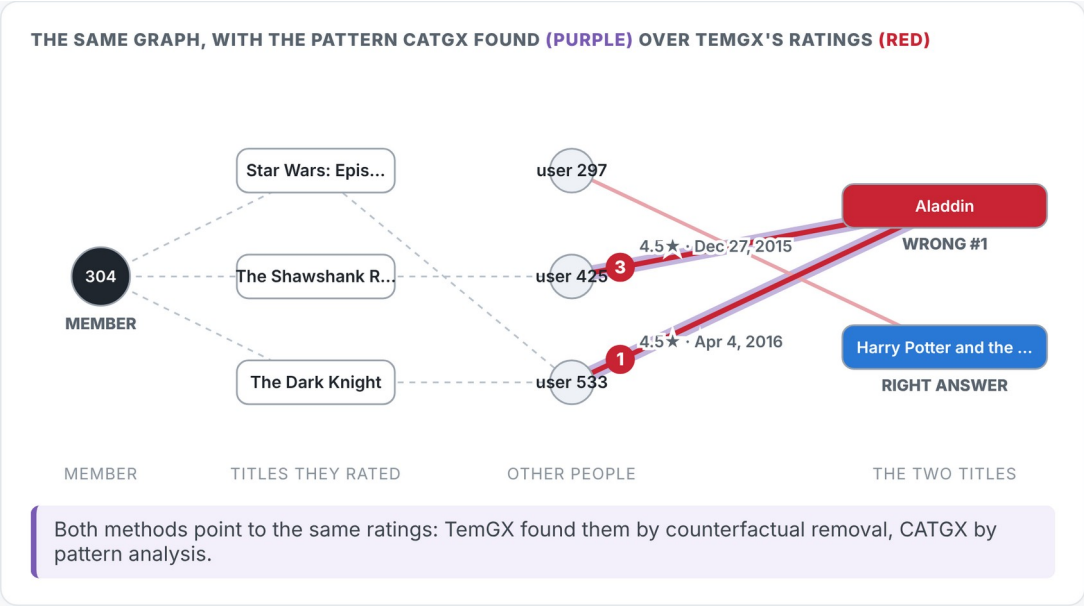
b: context cell (same recency, duration, activity)
ω(b): how common the cell is
μ − μbase: removing the pattern vs a comparable one in b

Member 304, shared taste: each removal costs 3.0 points of the history model's preference vs 0.9 for an equally recent, equally active pattern.

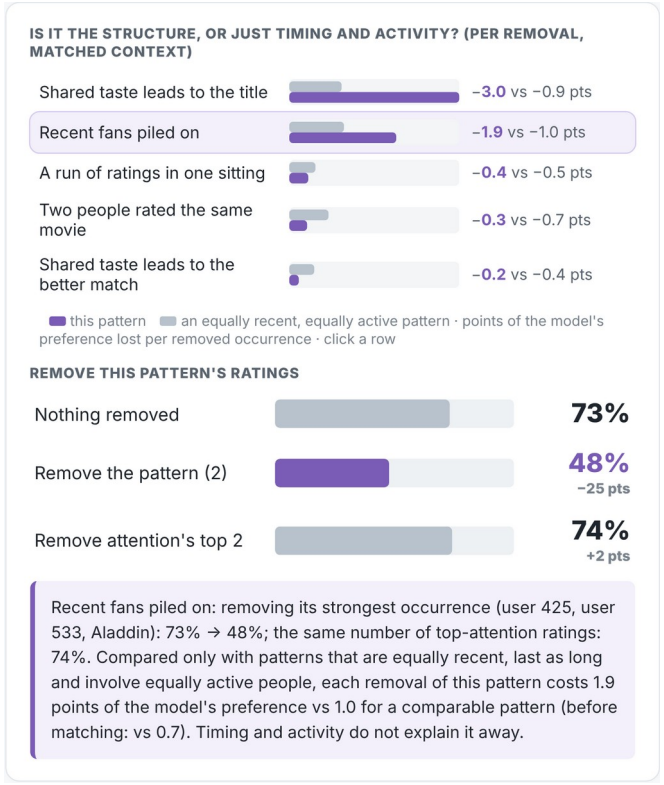
1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[2] Lu et al., KDD 2026, Fig. 3 and §6, Algorithm 2. [7] Pearl, 2009, adjustment background.

Two Aladdin patterns survive matching on timing and activity



Purple: the pattern CATGX found; **red:** TemGX's ratings. Both point to the same recent Aladdin fans.



73% → 48%

removing the two recent-fan ratings (attention: 74%)

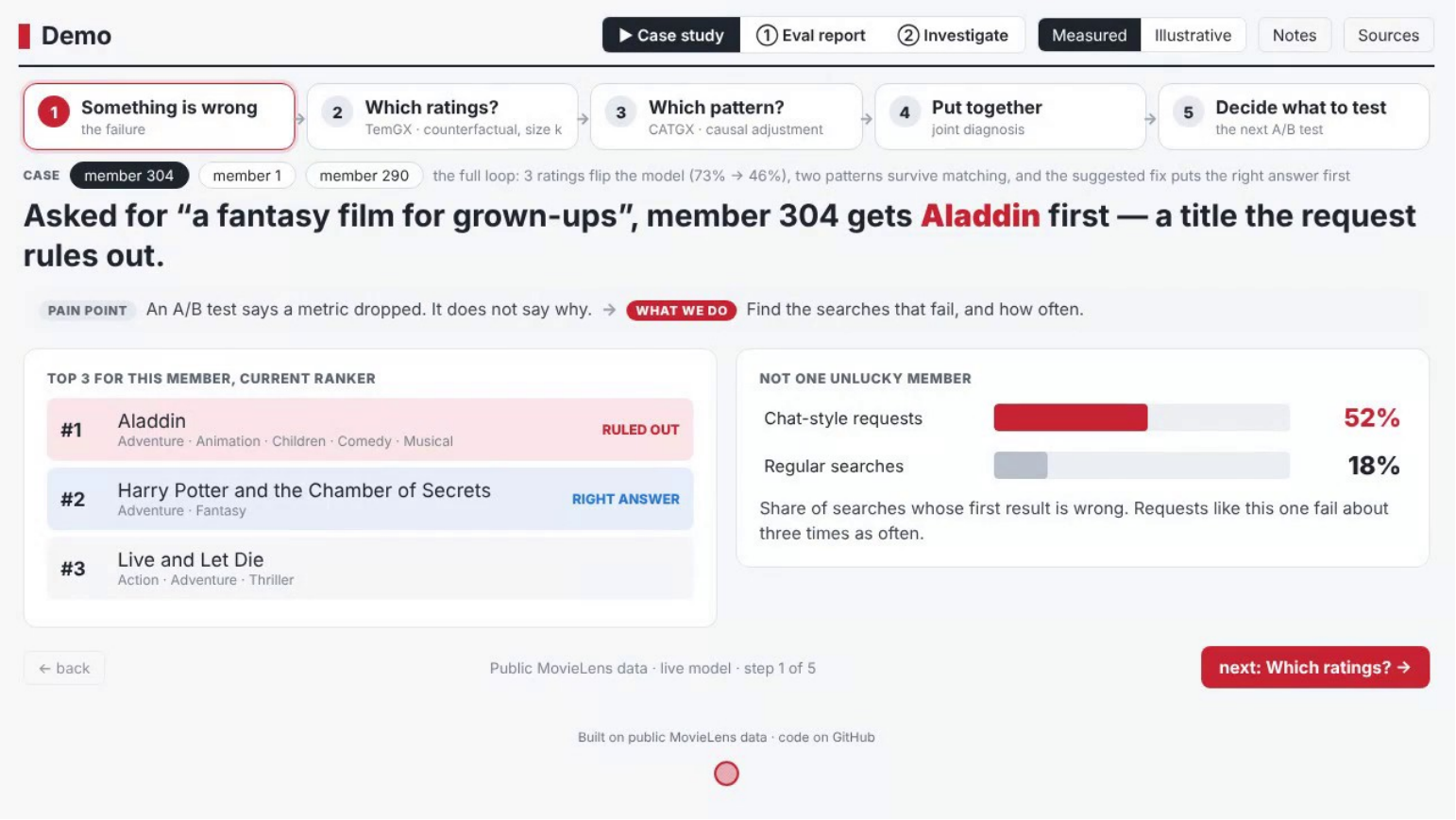
Shared taste: 73% → 62% (attention: 74%)

Both survive matching on timing and activity

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

[2] Lu et al., KDD 2026. Demo on MovieLens [5], member 304.

Aladdin comes first; the right answer is only second



PAIN POINT
An A/B test says a metric dropped. It does not say why.

WHAT WE DO
Find the searches that fail and how often.

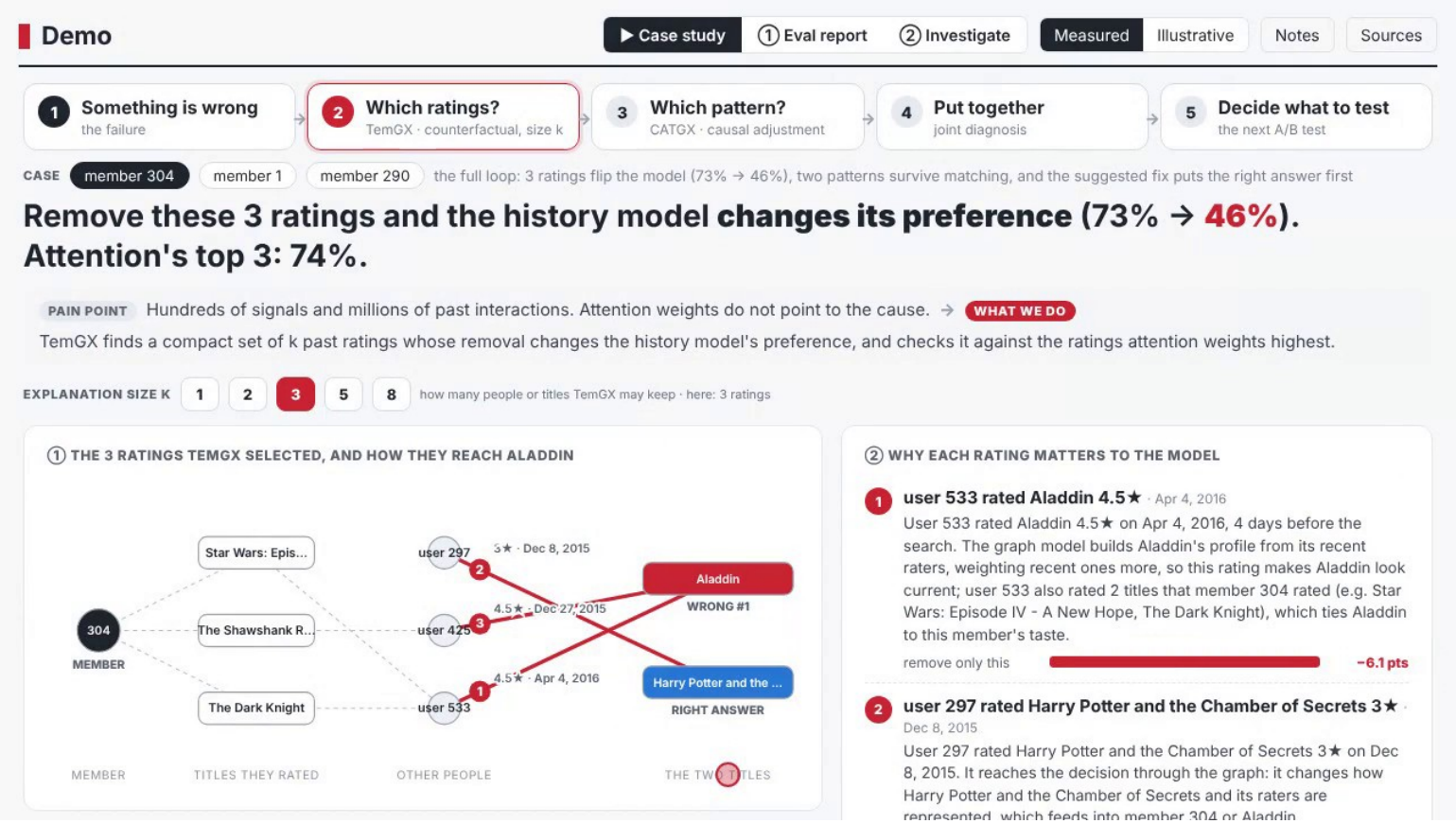
WHAT WE SEE
Member 304: Aladdin (an animated family film) #1, Harry Potter and the Chamber of Secrets #2. Chat-style requests: 52% start with a wrong title, vs 18%.

Try it yourself
Code, models and data:
github.com/nicej1899/search-ranking-investigation-lab

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

TemGX finds and tests the ratings the model depends on



PAIN POINT
Hundreds of signals; attention weights do not point to the cause.

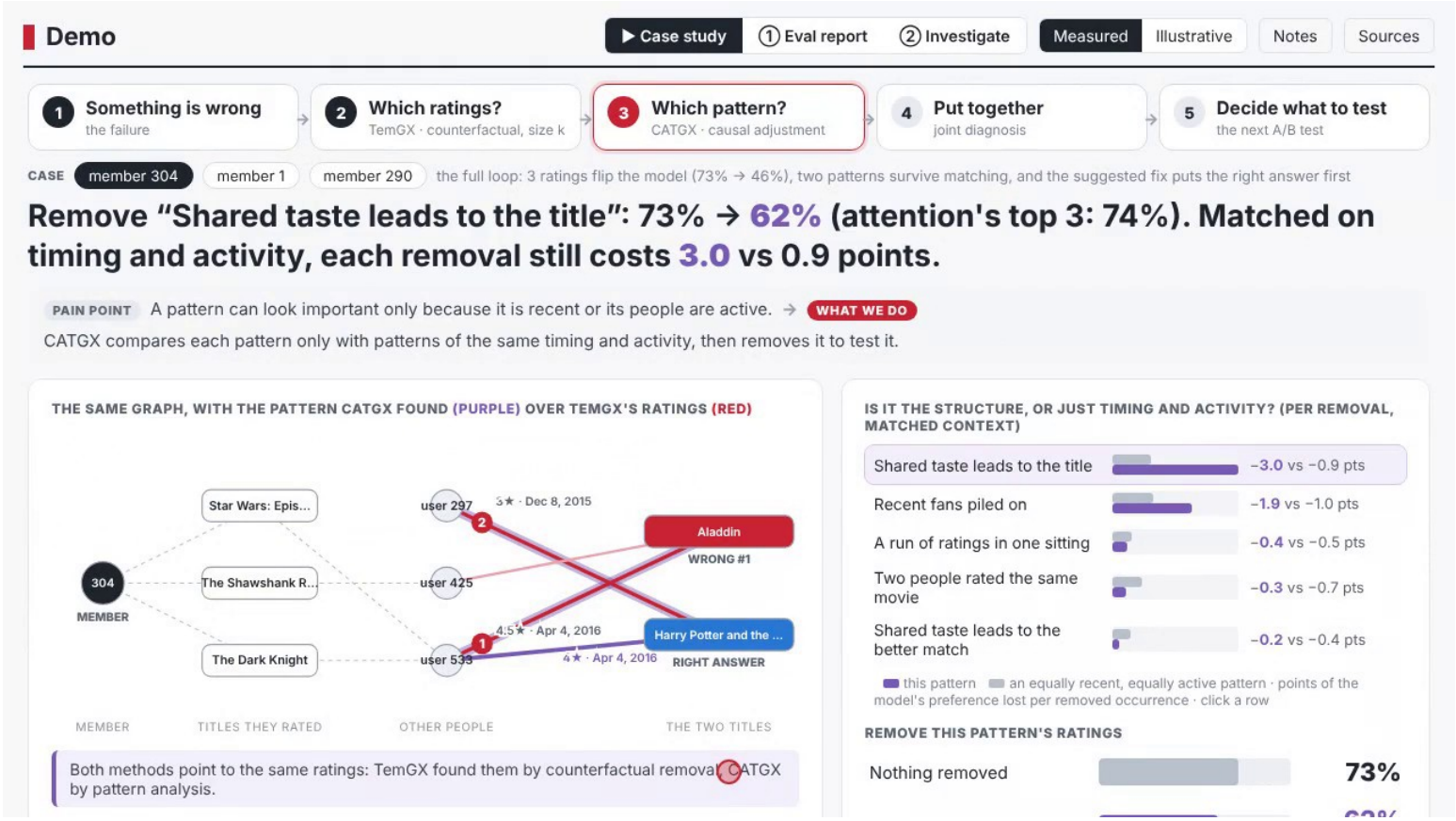
WHAT WE DO
Select a size-k explanation, then remove it and run re-inference on the same model; compare with the ratings attention weights highest.

WHAT WE SEE
k = 3: 73% → 46%, the history model's preference flips; attention's top 3: 74%. Across 60 failed searches with this request, TemGX beats attention in 52.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

CATGX checks whether a pattern survives matching



PAIN POINT
A pattern can look important only because it is recent or its people are active.

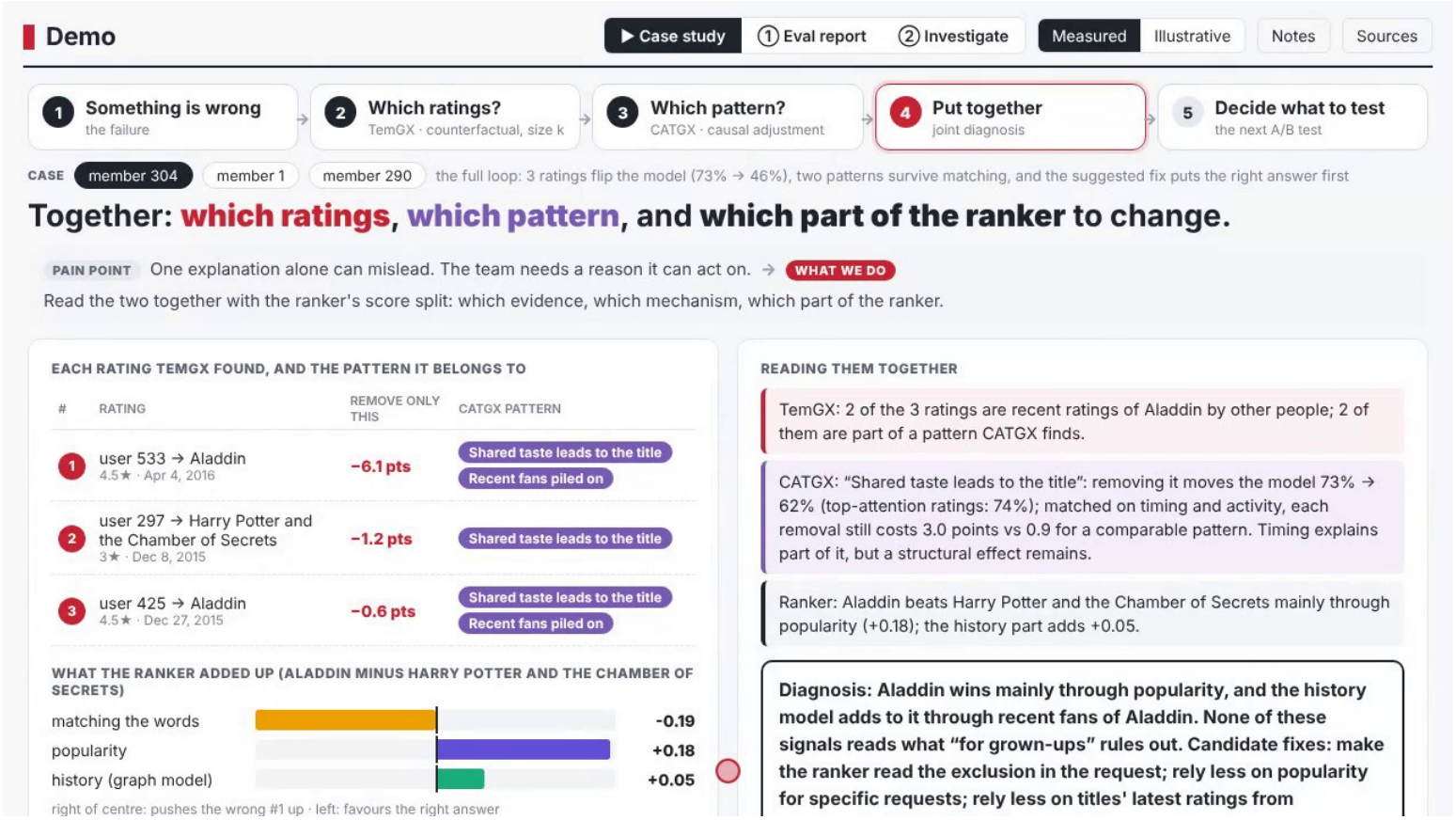
WHAT WE DO
Compare each pattern only with patterns of the same recency, duration and activity.

WHAT WE SEE
Removing shared taste: 73% → 62%; the recent-fan pair: 73% → 48%; attention of the same size: 74%. Both survive matching on timing and activity.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

Which ratings, which pattern, which part of the ranker



PAIN POINT
One explanation alone can mislead. The team needs a reason it can act on.

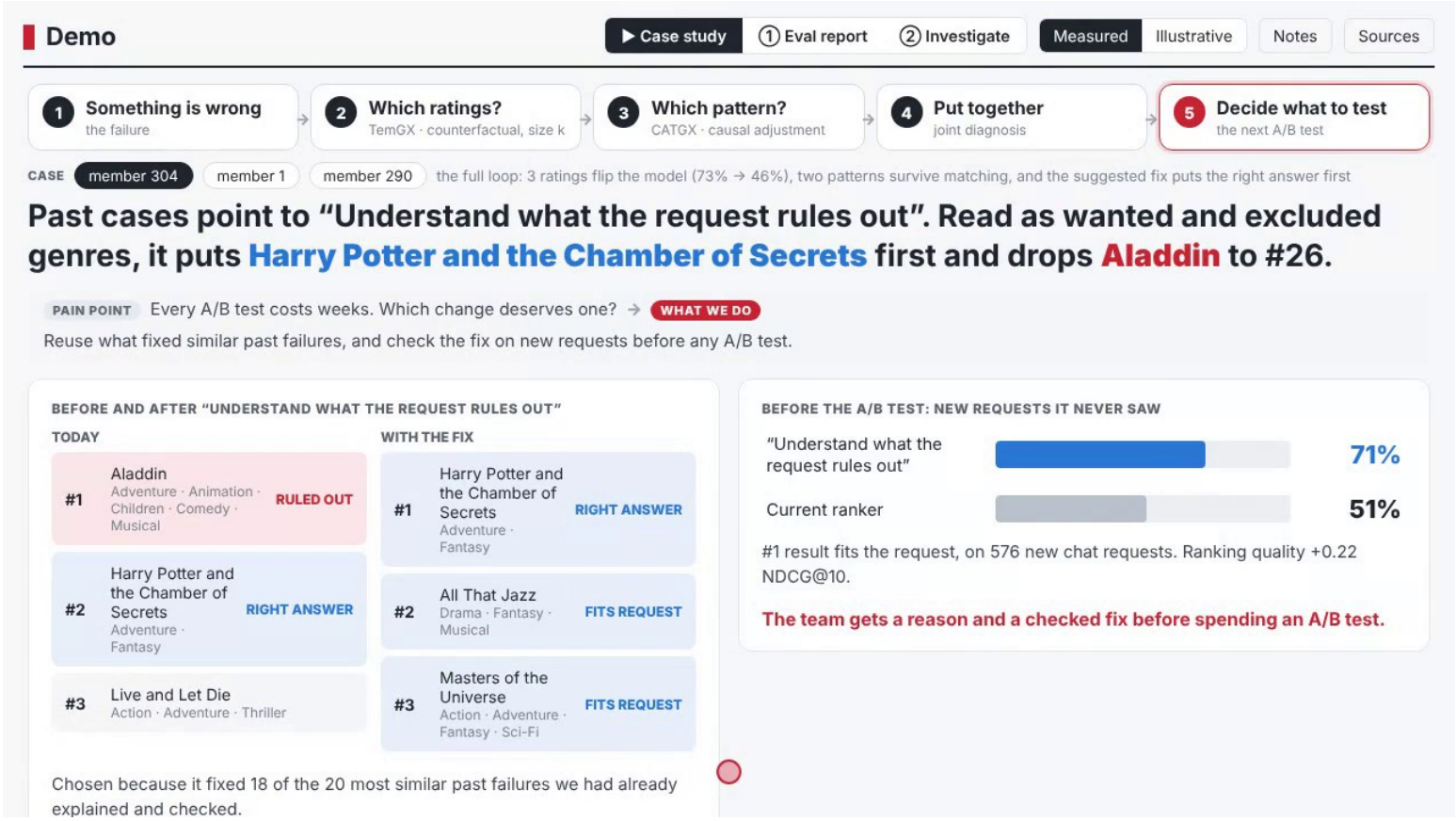
WHAT WE DO
Read TemGX, CATGX and the ranker's three signals together.

WHAT WE SEE
Aladdin wins on popularity (+0.18) and history (+0.05, its recent fans); text match already prefers Harry Potter (−0.19). Nothing reads what "for grown-ups" rules out.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

A checked fix before spending an A/B test



PAIN POINT
Every A/B test costs weeks. Which change deserves one?

WHAT WE DO
Reuse what fixed similar verified failures; check the fix on new requests.

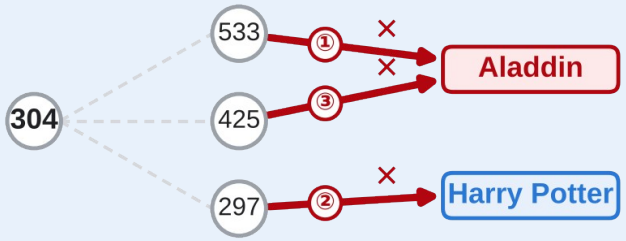
WHAT WE SEE
Recommended: understand what the request rules out (18 of 20 similar cases). Read as wanted and excluded genres: Harry Potter → #1, top 5 all fantasy; Aladdin → #26. New chat requests: #1 fits 51% → 71%.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Dataset: MovieLens [5]. Demo.

Evidence tells us where to steer the ranker, failure by failure

Event level single ratings

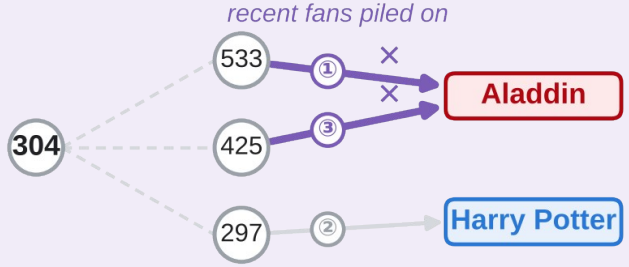


Change drop ①②③ for this member only

304 **Harry Potter #1, Aladdin #2**

At scale flips 10 of 60 failures (history leads in 2)

Pattern level this member's pattern



Change drop only ① ③, the "recent fans" pattern

304 **Harry Potter #1, Aladdin #2**

At scale same pattern as a global rule: Aladdin stays #1

Request level every request like this

"a fantasy film for grown-ups"

wants: Fantasy

rules out: Animation, Children

Aladdin #1 → #26

Animation · Children

Harry Potter → #1

top 5 all fantasy, none animated

Change read what the request wants and rules out

304 **Harry Potter #1, Aladdin #26**

At scale new chat requests: #1 fits 51% → 71%

Personalized repair: each failure gets the lever its own evidence points to.

Choosing it from the most similar verified failures fixes 39% of 280 held-out failures, vs 24% for one global fix (341 verified diagnoses).

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Demo on MovieLens [5], member 304 (TGAT-2 history model). At scale: 60 failed searches for this request; 576 new chat requests (the request-level fix reads genres, as the relevance labels do); 341 DEV diagnoses, 280 held-out failures, paired gain +15.0 pp, 95% CI [+10.4, +19.6].

From explaining one failure to steering the ranker

Richer events

Multimodal and cross-platform actions joined into one event, e.g. trailer → soundtrack → rating.

Already measured: on stand-in semantic IDs, Animation / Children codes are 22.4% of wrong #1 results vs 13.3% of candidates.

From explanation to intervention

Remove, down-weight or inject the evidence to steer the ranker, then test it offline before an A/B test, as shown for member 304.

Already measured: for member 304, dropping ① and ③ puts Harry Potter #1.

Transfer across surfaces

Does evidence found on one surface, such as search, predict behaviour on another, such as the home page or games?

Already measured: ignoring strangers' latest ratings (an idea from the explanations) lifts held-out new-member NDCG@10 by +0.014.

Multi-agent diagnosis

An event agent, a causal-analysis agent and an explanation agent run the five steps on every new failure slice.

Already measured: 1,726 regression checks built from verified failures, ready for a coding agent to run.

Goal: every failed search comes with a checked reason and a fix worth testing

[1] Lu et al., ICLR 2026. [2] Lu et al., KDD 2026. Code and Demo: github.com/nicej1899/search-ranking-investigation-lab

Backup guide: choose the question being asked

01 Definitions and causal meaning

02 Reading a temporal graph

03 Fixed-model intervention protocol

04 Budgets and the guarantee

05 Causal target versus adjusted score

06 Confounders, mediators and assumptions

07 A numerical adjustment example

08 Motif library and codebooks

09 HNSW and exact verification

10 Sampling weights and support

11 Deletion controls and budgets

12 Evaluation populations

13 Planted-effect metric

14 Attention and retrieval comparisons

15 Diagnosis retrieval and repair

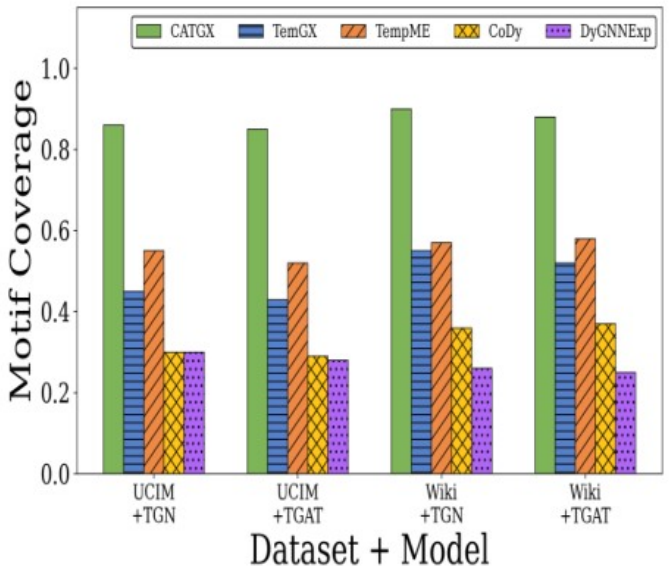
16 Semantic IDs and cold start

17 Runtime and deployment

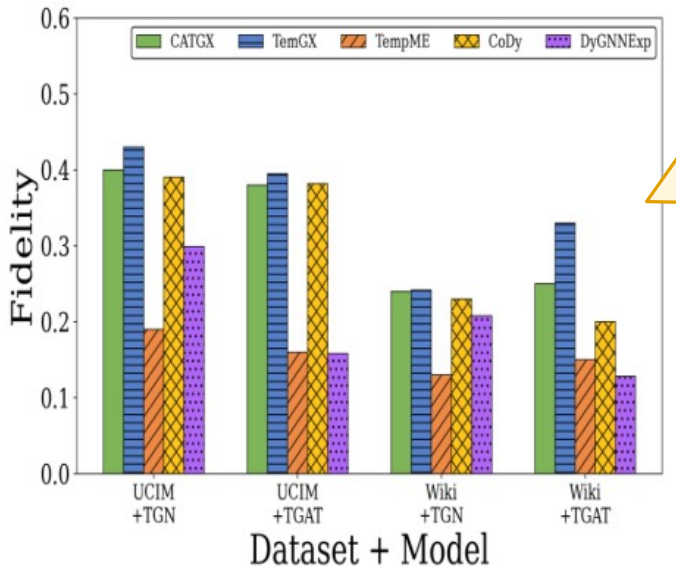
18 My contribution and limitations

Published references are collected on the final page.

CATGX: motif coverage and fidelity



(a) Overall Motif Coverage



(b) Overall Fidelity Comparison

CATGX recovers far more complete motifs (coverage ≈ 0.85–0.9 vs ≤ 0.6), at fidelity close to the strongest baseline

Data: Wikipedia edits / UCIM messages. Models: TGN / TGAT. Baselines: TemGX, TempME, CoDy, DyGNNExp.

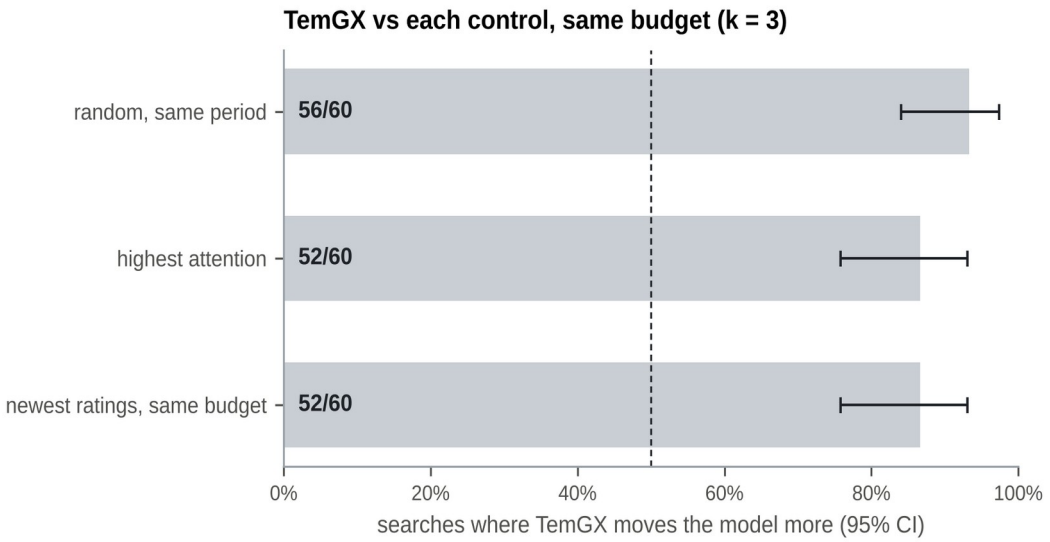
200 test predictions, k = 50, aligned budgets. Coverage counts complete motifs; fidelity tests model-output behaviour.

[2] Lu et al., KDD 2026, §7, Fig. 5(a–b). Original coverage and fidelity results.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Evidence must beat controls and recover a known effect

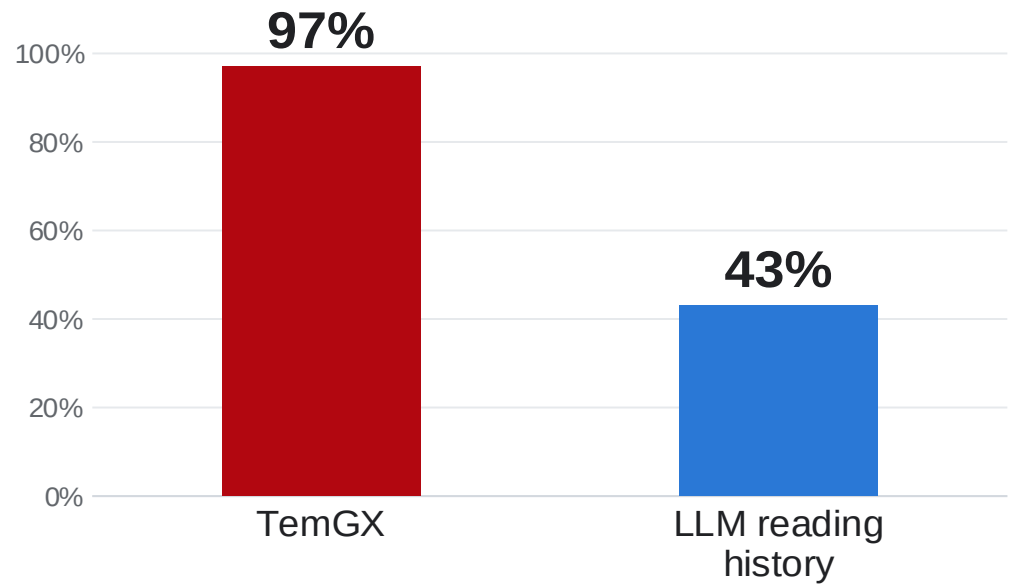
Deletion controls



52 / 60 beat attention's top 3

Each bar: share of failed searches where removing TemGX's 3 ratings moves the model more than removing 3 ratings picked by that control.

Known planted mechanism



97% vs 43% of the effect removed

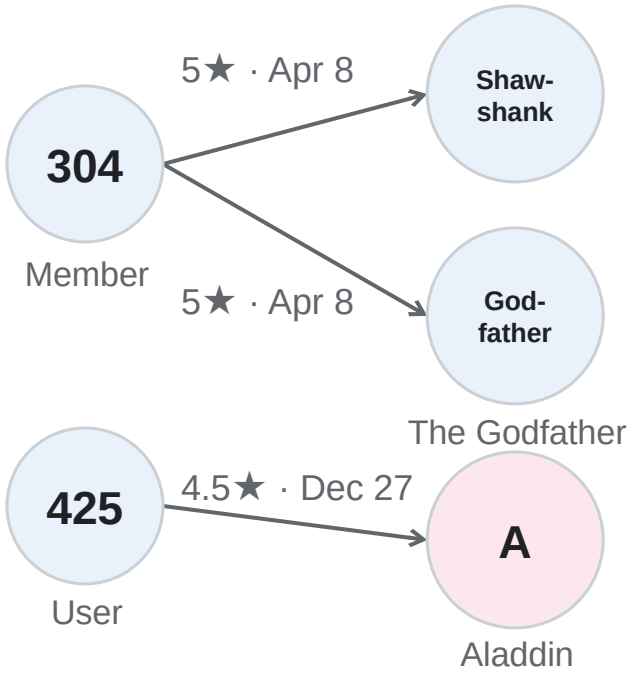
We plant a known pattern in the data, retrain, and measure how much of its effect each method's picks remove. LLM: a language model reading the history as text.

1 Failure
2 Events TemGX
3 Patterns CATGX
4 Together
5 Fix

Demo on MovieLens [5]. Left: 60 failed searches for “a fantasy film for grown-ups”, k = 3, Wilson 95% CIs. Right: 21 probes, median planted effect removed.

From timestamped history to a model prediction

Ghist: history observed before time _t_



$z = fE(Ghist)$ $y = fD(z)$

1 Encode

_fE_ Trained encoder

Summarizes temporal interactions

2 Represent

$z = [z_1, z_2, \ldots]$

Learned numerical representation

3 Decode

_fD_ Target score _y_

Example target: Aladdin versus Harry Potter and the Chamber of Secrets

History excerpt: ratings have values and timestamps. The full history goes into the same fixed model.

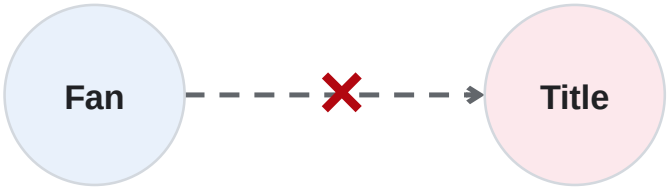
[1] Lu et al., ICLR 2026, §2. Example history: Demo on MovieLens [5], member 304.

One removal measures local sensitivity

Before



After removing one rating



Model probability: 0.80

Model probability: 0.70

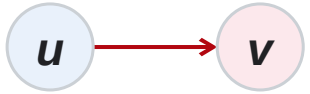
Influence = |0.80 − 0.70| = 0.10

[1] Lu et al., ICLR 2026, §3. [6] Wachter et al., 2018. Numerical values are a teaching example.

A node's influence is weighted by resistance and age

$$c(u,\tau) = \inf(u,v,\tau) \times g(\mathrm{TRD}(u,v,\tau)) \times \exp[-\lambda(t-\tau)]$$

u = historical node v = target τ = event time t = prediction time

	Influence	Resistance gate	Recency weight	Contribution
	 Temporal paths carry influence	 Higher resistance gives a smaller g	 Older evidence receives less weight	
Toy factors at one timestamp				
u_1	0.28	× 0.80	× 0.90	= 0.2016
u_2	0.28	× 0.40	× 0.50	= 0.0560

Average c over the time window, then sum over selected nodes to obtain $\Phi(V_s,v)$.

[1] Lu et al., ICLR 2026, §3. Temporal resistance and time decay. Toy factors, not experiment results.

Selection keeps explanatory nodes and their temporal connectors

1 Form candidates

Use *L*-hop temporal neighbours inside each sliding window.

2 Select and verify

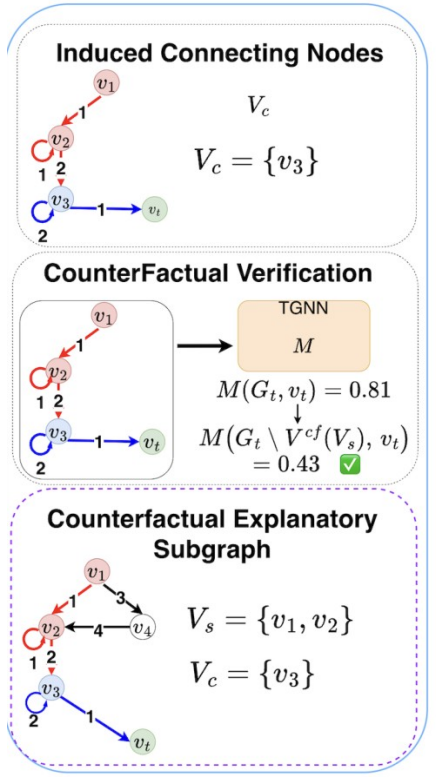
Rank by Φ, test candidate evidence with the fixed TGNN, and update the selected set.

3 Preserve temporal paths

Add connector nodes *Vc* and induce the explanatory subgraph.

4 Continue across windows

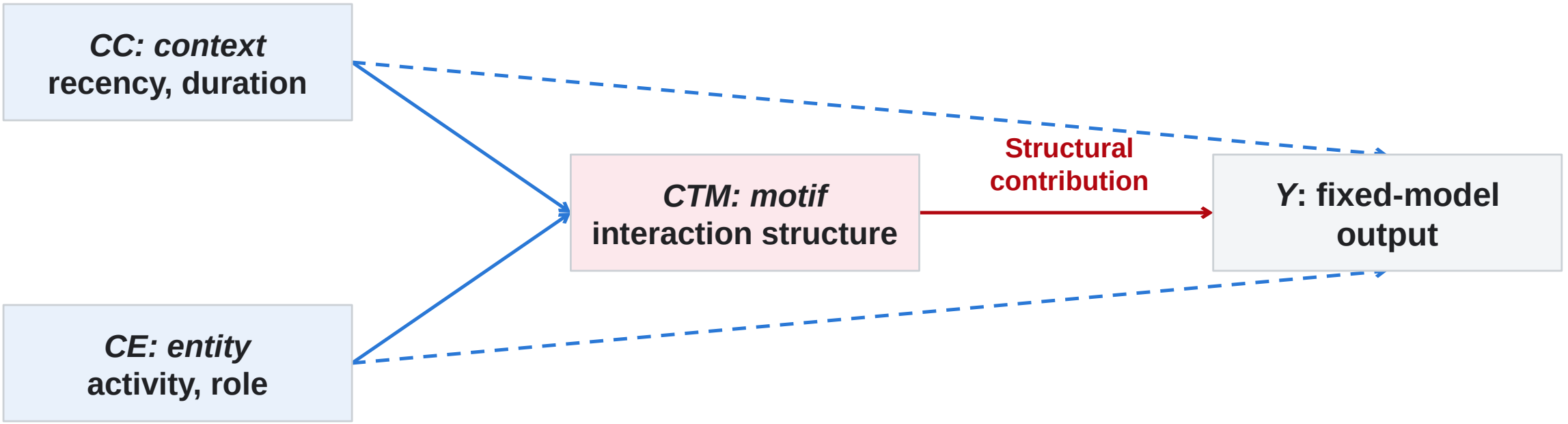
Update the explanation as historical interactions evolve.



The paper gives a (1−1/e) guarantee for its defined influence objective under the stated assumptions.

[1] Lu et al., ICLR 2026, §4 and Lemmas 1–3. Guarantee concerns the stated influence objective and assumptions.

Three codebooks separate structure, context and entity behaviour

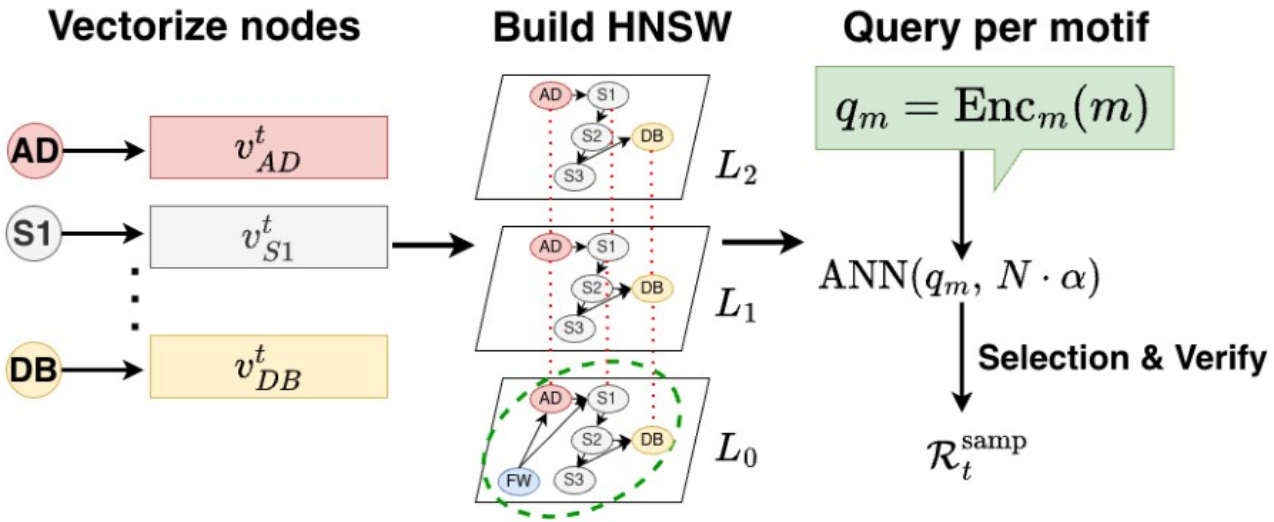


Movie example: compare rating patterns at similar times and activity levels

Target: dependence of the model output, conditional on the stated causal assumptions.

[2] Lu et al., KDD 2026, §3, Eq. (1), Fig. 2. [7] Pearl, 2009. Arrows specify the assumed model, not an identified effect on member behaviour.

Verified retrieval and sampling make motif selection practical



Verify candidates

Check node and edge labels and temporal order in the original graph.

Stratify and weight

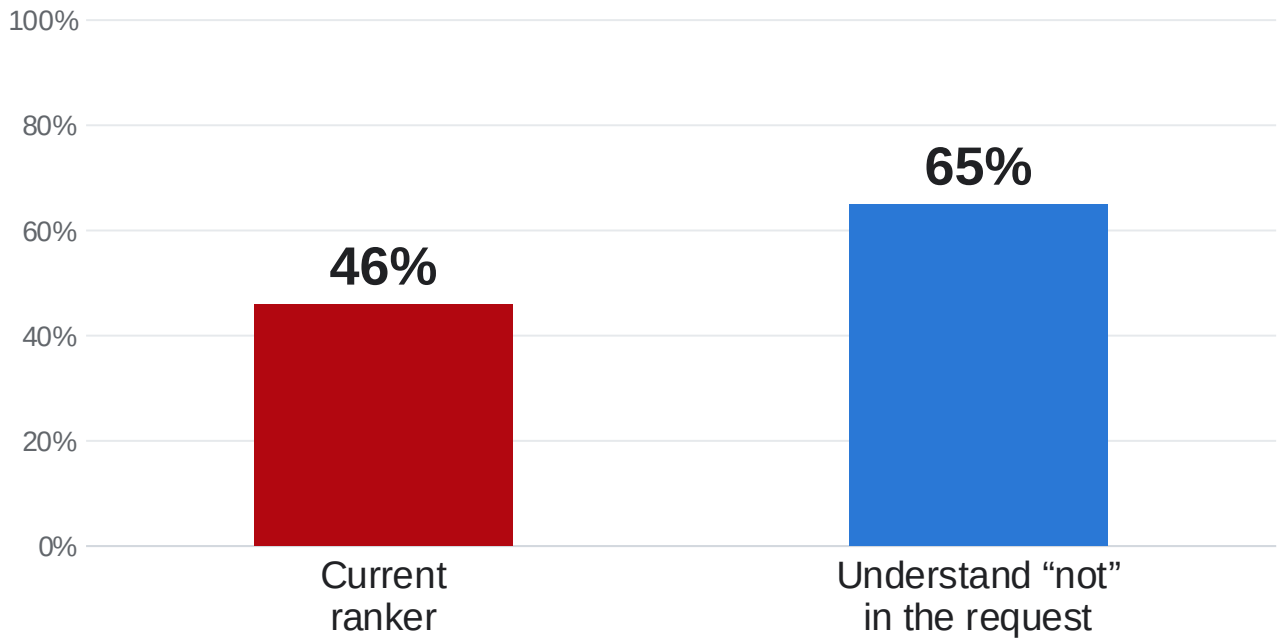
Group by recency, duration and activity. Use $w(r) = 1/\pi(r)$.

Occurrence selection: rank by $w(r)g(r;M) / |Er|$ and retain complete motifs under budget.

Full-population unbiasedness requires exact retrieval and positive support. ANN limits the covered population.

[2] Lu et al., KDD 2026, §§5–6. [11] Malkov & Yashunin, IEEE TPAMI, 2020. Exact-retrieval and positive-support assumptions govern unbiased totals.

A better constraint check can still carry a quality cost



65% vs 46%

excluded title kept out of the top three

−0.016

change in NDCG@10

Measure the repair’s benefit and its ranking-quality tradeoff

Experiment: Demo. Compliance: 576 held-out chat requests. Ranking quality: 468 requests, member bootstrap. Dataset: [5]. Evaluation background: [3,4].

Explanation, counterfactual and causal inference answer different questions

Explanation

Which observed inputs does this fixed predictor depend on?

Counterfactual

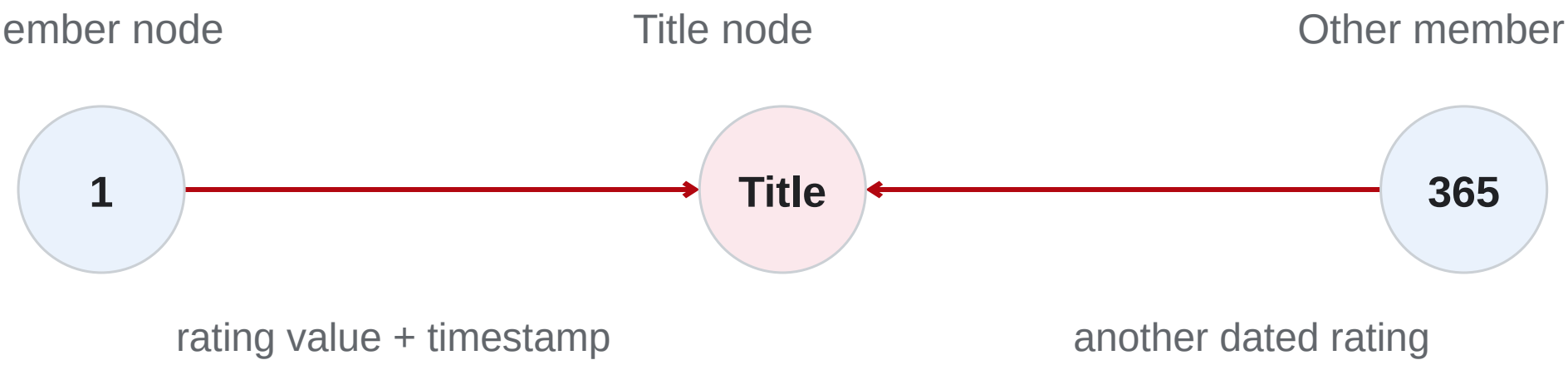
Would a permitted change to those inputs alter its decision?

Causal interpretation

Which dependence remains after adjustment under an explicit causal model?

[1,2,6,7] Published sources listed on the final References page.

A temporal graph records who interacted, with what, and when



Temporal neighbourhood + fixed prediction target + history cutoff

[1,5,8] Published sources listed on the final References page.

The intervention protocol holds the trained predictor fixed

Allowed change

Remove the selected historical events, keeping the target and model parameters fixed.

Required state handling

Recompute the affected representation from the permitted history. Stateful temporal models require consistent replay.

Leakage check

Use only events before the target time. Match the intervention and its controls.

[1,2] Published sources listed on the final References page.

The paper budget, Demo budget and approximation guarantee are distinct

TemGX formulation

k counts selected explanatory nodes; required connector nodes preserve temporal reachability.

Demo deletion test

Most controls remove three rating events. The CATGX case contains a two-rating occurrence.

Guarantee

The (1−1/e) bound applies to the defined influence objective under the paper’s assumptions. It is not a universal minimal-flip guarantee.

[1,2] Published sources listed on the final References page.

The causal estimand and the implemented adjusted score are different

Causal target under the assumed graph

$$P(Y \mid do(M = m)) = \sum c \; P(Y \mid M = m, C = c) \; P(C = c)$$

Implemented CATGX score

$$S(m) = \sum b \; \omega(b) \; [\mu(m,b) - \mu base(b)]$$

M: motif structure. C: measured context/activity. Y: fixed-model output.

S(m) is an adjusted score; the Demo reports the same comparison as points lost per removal (member 304: 3.0 vs 0.9).

[2,7] Published sources listed on the final References page.

Adjustment requires measured common causes and comparable support

Common cause

A pre-existing condition affects motif occurrence and the model output. Timing or activity may play that role.

Mediator or consequence

Do not automatically adjust for a variable produced by the motif itself. Measurement timing matters.

Assumptions to check

Causal graph, observed confounding, consistency and overlap. Unmeasured factors or unsupported cells weaken the interpretation.

[2,7] Published sources listed on the final References page.

A teaching example shows how context can inflate a raw difference

Context	Pattern	Count	Mean score
Recent / active	A	80	0.80
Recent / active	B	20	0.75
Older / less active	A	20	0.40
Older / less active	B	80	0.35

Raw: A 0.72 vs B 0.43 (difference 0.29)

Same 50/50 context mix: A 0.60 vs B 0.55 (difference 0.05)

[2,7] Published sources listed on the final References page. Numbers are a teaching example, not experimental data.

The motif library defines which structural hypotheses can be tested

Template

A small labelled interaction graph with an event order. The library supplies candidate hypotheses.

Occurrence

A mapping to actual nodes and edges that satisfies labels, connections and temporal constraints.

Codebooks

CTM records structure; CC records timing; CE records participant properties. A dashed context link is not an extra rating edge.

[2,8] Published sources listed on the final References page.

HNSW retrieves candidates; graph verification establishes valid occurrences

Fast candidate retrieval

Use embeddings and an HNSW index to narrow the search. Nearest-neighbour retrieval is approximate.

Exact validity checks

Verify labels, required graph connections and event order against the original temporal graph.

Remaining limitation

Verification removes invalid candidates. It does not recover valid occurrences that approximate retrieval never returned.

[2,11] Published sources listed on the final References page.

Inverse-probability weights correct sampling within supported populations

Sample probability π(r) = 0.10

Weight w(r) = 10

Sample probability π(r) = 0.50

Weight w(r) = 2

$$w(r) = 1 / \pi(r)$$

Exact retrieval + positive support are required for the unbiased-total result.
Weights do not recover occurrences that retrieval never exposed.

[2] Published sources listed on the final References page.

Faithfulness tests compare input changes with clearly defined controls

Same target

Run re-inference on the same graph component for the same pair preference after removal. Report direction and magnitude.

Control budget

Random, attention and recency edge controls remove three ratings. Report CATGX’s two-rating occurrence separately.

Interpretation

52/60 (against attention) is an observed paired win count, not population-wide certainty. The 32-search study of a second request (backup) shows the same direction.

Methods: [1,2,9,10]. Experiment: Demo, 60 deletion cases (member 304's request) and 32 (a second request); CATGX and edge-control budgets differ.

The evaluation populations answer different questions

Question	Evaluation population
Where does the ranker fail?	Development: chat 636; regular 848
Does the selected evidence matter?	32 deletion cases; 21 planted probes / 2 patterns
Can diagnosis choose a repair?	343 development diagnoses; 280 held-out failures
Does a repair improve the intended metric?	Compliance: 576 chat requests; quality: 468 requests
Does the cold-start intervention help?	528 new-member requests

All counts describe Demo; the paper benchmarks are separate.

Dataset: MovieLens [5]. Counts: Demo experiment populations, separate from paper benchmarks.

97% measures planted effect removed, not cause-identification accuracy

Known reference

A controlled experiment embeds a specified mechanism, providing a reference contribution to remove.

Reported result

Median fraction of planted effect removed: TemGX 97%, LLM reading history 43%; 21 probes, 2 patterns.

Limit

This checks a controlled mechanism. It does not establish the unique real-world cause of every natural failure.

Method: [1]. Experiment: Demo, 21 probes and 2 planted mechanisms. These percentages are Demo results.

Attention and retrieval require tests suited to their intended use

Attention

A high weight may suggest evidence, but its explanatory meaning depends on the model and the test.

Retrieved context

Similarity or recency helps supply relevant context. It does not by itself demonstrate dependence of a target prediction.

Operational check

Remove selected inputs and compare with controls. The cited attention debate concerns NLP models, not a universal theorem about temporal graphs.

[9,10,1] Published sources listed on the final References page.

Diagnosis retrieval proposes a repair; held-out checks evaluate it

Input and output

Input: a new diagnosed failure. Retrieve similar checked cases and propose a repair using their measured outcomes.

Repair lookup result

341 development diagnoses, 280 held-out failures; working repair selection 39% versus 24% for one global fix.

Separate policy test

“Understand what the request rules out” raises exclusion compliance to 65% versus 46%, with −0.016 NDCG@10. An online test remains separate.

Background: [3,4,5]. Results: Demo offline diagnosis-retrieval and repair experiments.

Semantic-ID and cold-start findings are specific offline starting points

Semantic-ID proxy

Residual k-means codes, not Netflix’s production codebook. Animation/Children codes: 22.9% of wrong-first results vs 13.5% of candidates.

Cold start

Ignoring each title’s latest stranger ratings gives new-member NDCG@10 +0.014; n = 528, CI [+0.010,+0.019].

Next investigation

Explain which historical inputs drive each slice, then test with the team’s representations, logs and success metrics.

Dataset: MovieLens [5]. Results: Demo. Semantic IDs are a residual k-means proxy.

Offline scalability evidence does not define a production latency target

Published stress test

On a separate synthetic graph with one million edges: CATGX 48.2 s; TemGX 254 s, under the reported paper setup.

Practical deployment

Begin with offline failure analysis or sampled regression checks; measure retrieval, encoding and verification costs separately.

Evidence to collect

Recall versus runtime, memory, model-query cost and stability across windows. Choose a serving architecture only after those measurements.

[2,11,3] Published sources listed on the final References page.

My contribution connects scalable search with testable explanations

CATGX

HNSW retrieval, validity checks, stratified sampling and weighted explanation statistics.

TemGX

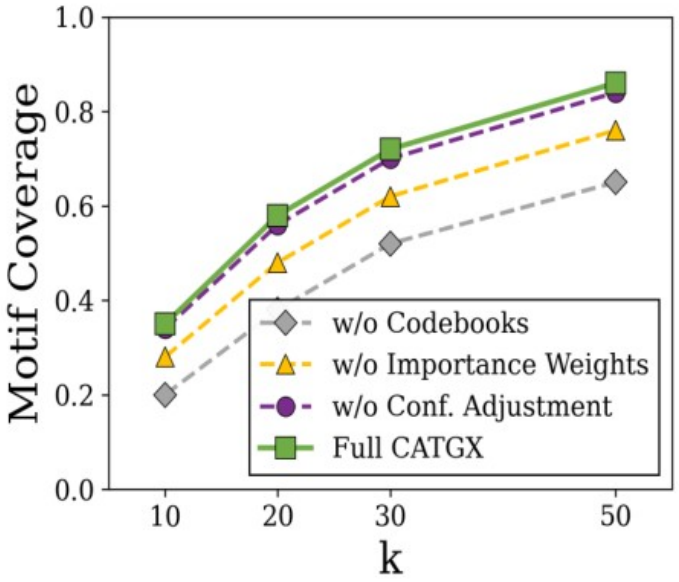
Temporal influence and resistance scoring, compact evidence selection and verification tooling.

Research judgement

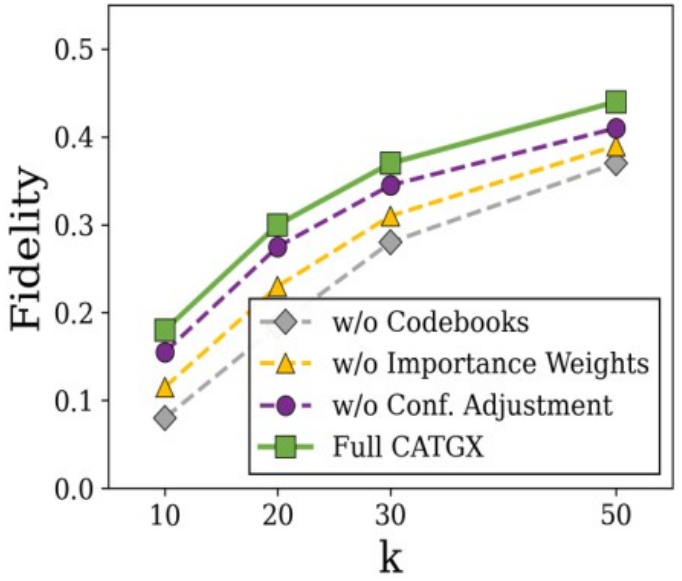
Separate model dependence from behavioural causality. Test failure slices and candidate repairs with explicit assumptions and held-out evidence.

[1,2,12] Published sources listed on the final References page.

Codebooks, weighting and adjustment improve the selected evidence



(e) Ablation k vs Coverage



(f) Ablation k vs Fidelity

1M-edge synthetic graph

Method	Runtime
CATGX	48.2 s
TemGX	254 s
TempME / CoDy / DyGNNExp	OOM

A separate stress test of computational cost

At *k* = 50, UCIM + TGN fidelity: full 0.44, no adjustment 0.41, no weights 0.39, no codebooks 0.37.

[2] Lu et al., KDD 2026, §7.2 and Appendix A.10. 1M edges is a separate synthetic offline runtime experiment.

The evaluation tests explanation quality across three prediction tasks

Task	Datasets and graph meaning	Predictive backbones
Traffic regression	METR-LA: 207 sensors PEMS-BAY: 325 sensors Roads and observed traffic speeds	STGCN, DCRNN
Link prediction	Wiki: user–page edits UCIM: timestamped messages	TGN, TGAT
Event classification	Multihost: audit events Elliptic++: Bitcoin transfers	TGN, TGAT

Controls Matched backbone, temporal neighbourhood, window and explanation budget.

Baselines TGNNExplainer, TempME, CoDy. TGVex for regression.

Measurements Fidelity / AUFSC, explanation size and runtime. Five independent seeds.

[1] Lu et al., ICLR 2026, §5 and Appendices D–F; Table 2.

The motif evaluation separates completeness, faithfulness and cost

Dataset	Interaction	Nodes	Events
Wiki	User–page edit, 172-dim features	9,227	157,474
UCIM	User-to-user message	1,899	59,835

Backbones and controls

TGN and TGAT. 70/15/15 train, validation and test splits. Main comparison: 200 test predictions, $k = 50$.

Baselines

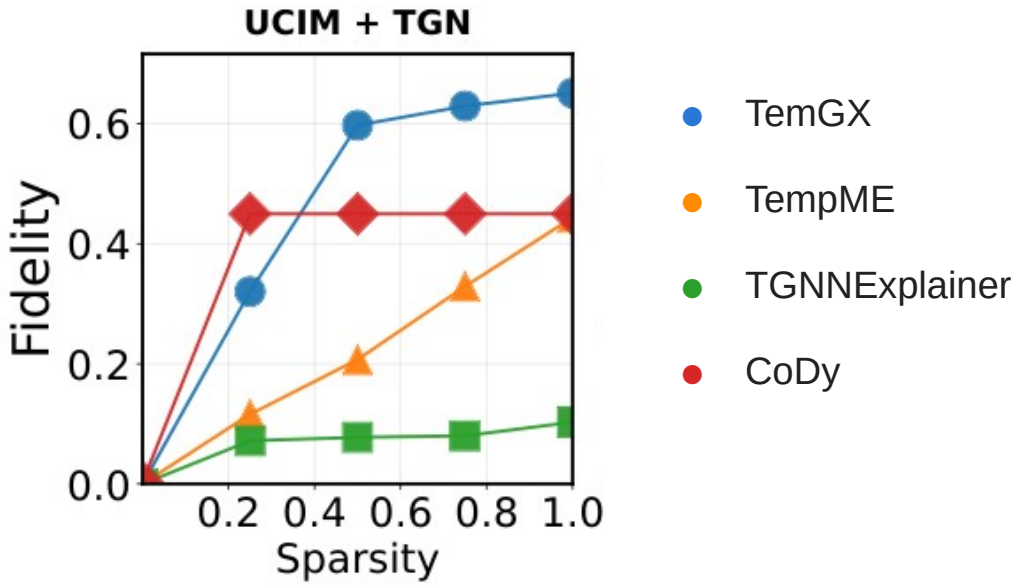
TemGX, TempME, CoDy and DyGNNExp, with aligned backbone, window and size settings.

Coverage counts complete motif occurrences. Fidelity measures model-output behaviour. Runtime is measured separately.

[2] Lu et al., KDD 2026, §7.1 and Appendix A.8.

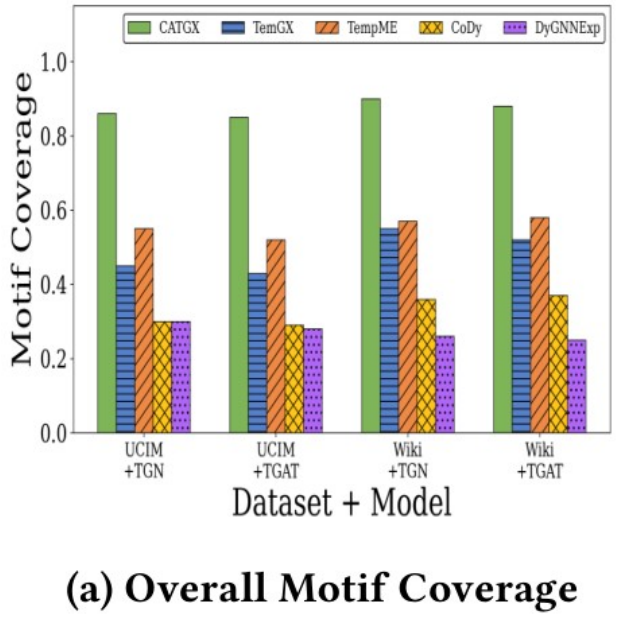
Published results support the explanation methods

TemGX / ICLR 2026



Strong fidelity across budgets

CATGX / KDD 2026

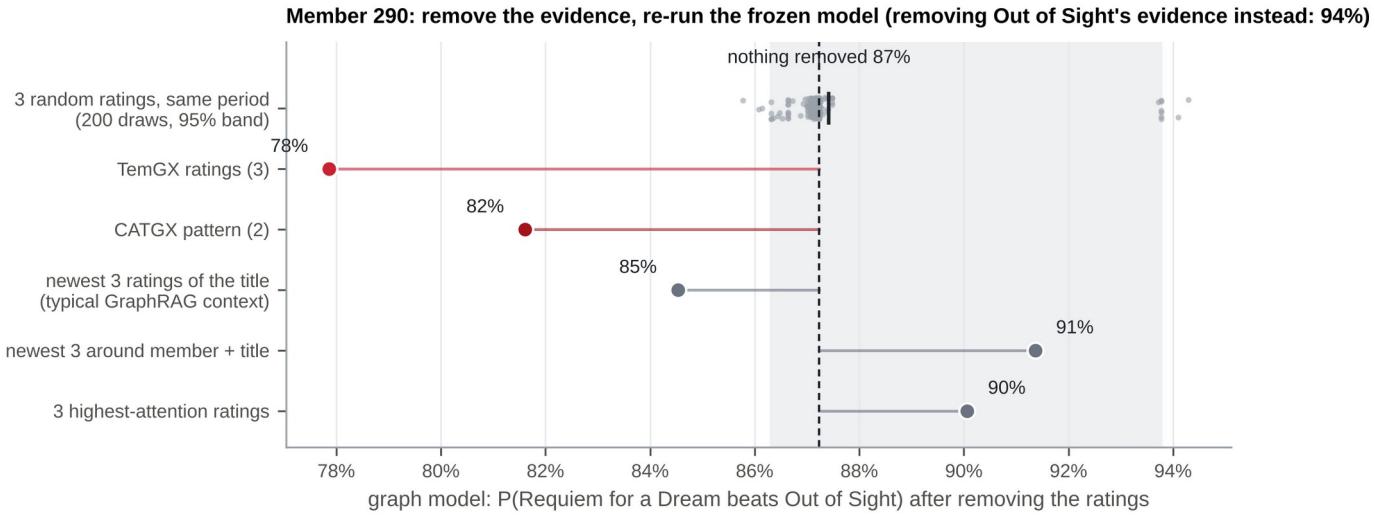


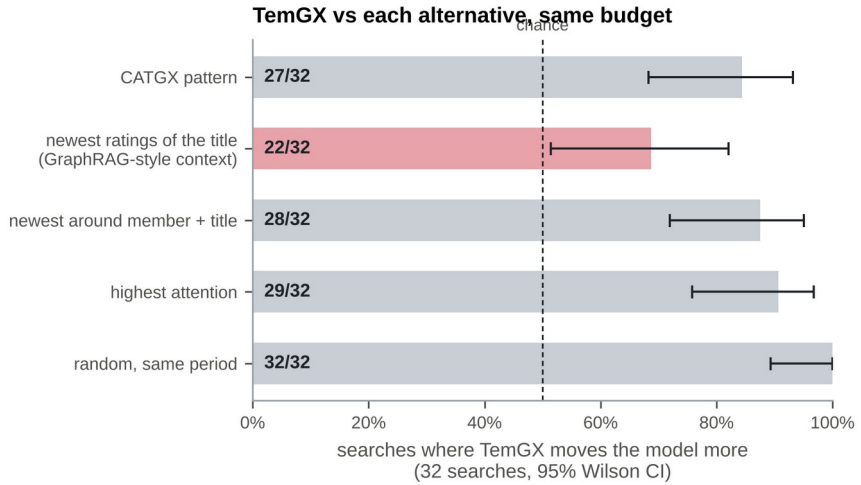
(a) Overall Motif Coverage

High coverage of complete motifs

[1] Lu et al., ICLR 2026, Fig. 4. [2] Lu et al., KDD 2026, Fig. 5(a). Original published panels.

TemGX outperforms the tested deletion controls



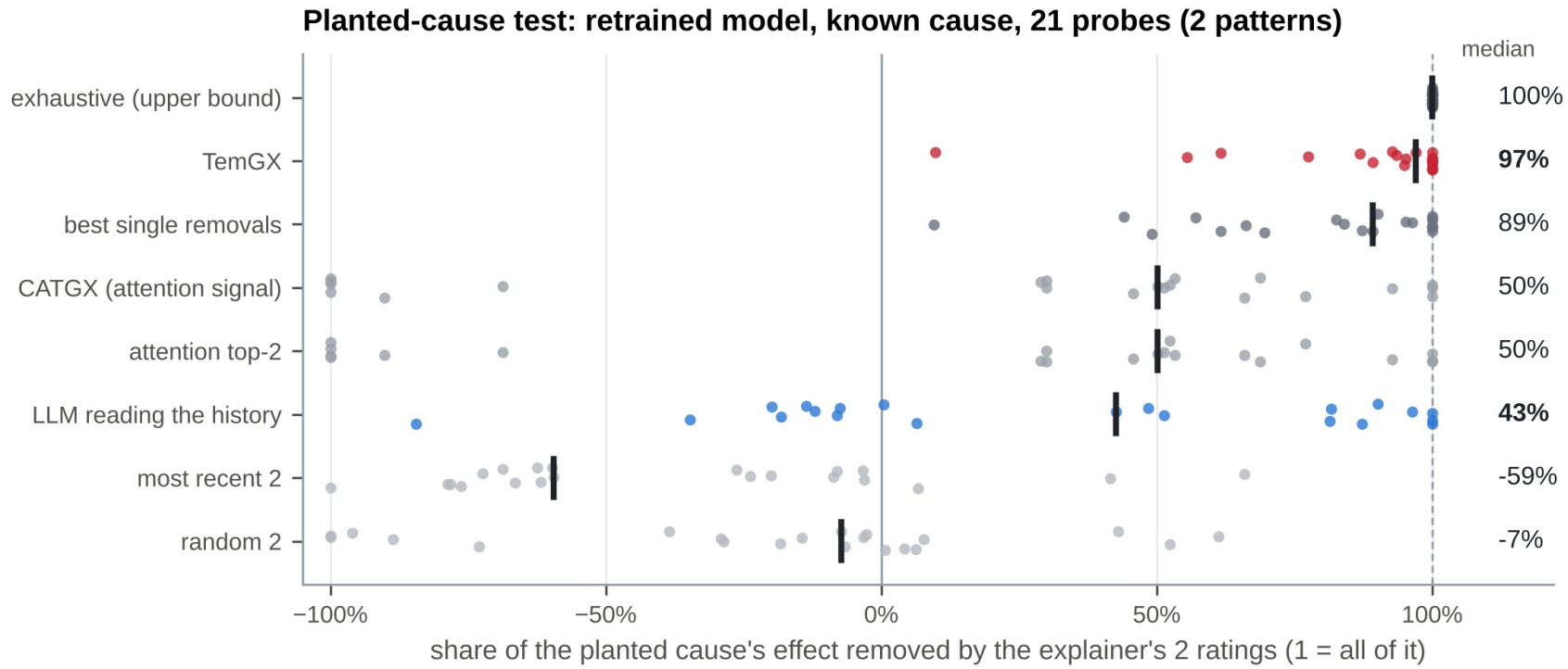


Second case, member 290: random band from 200 draws. Across 32 searches: Wilson 95% intervals. CATGX removes 2 ratings, edge controls 3.

What this gives your team: historical inputs tested against equal-size alternatives.

Method: [1]. Experiment: Demo on MovieLens [5]. CATGX removes 2 ratings; other edge controls remove 3.

TemGX removes 97% of the planted effect

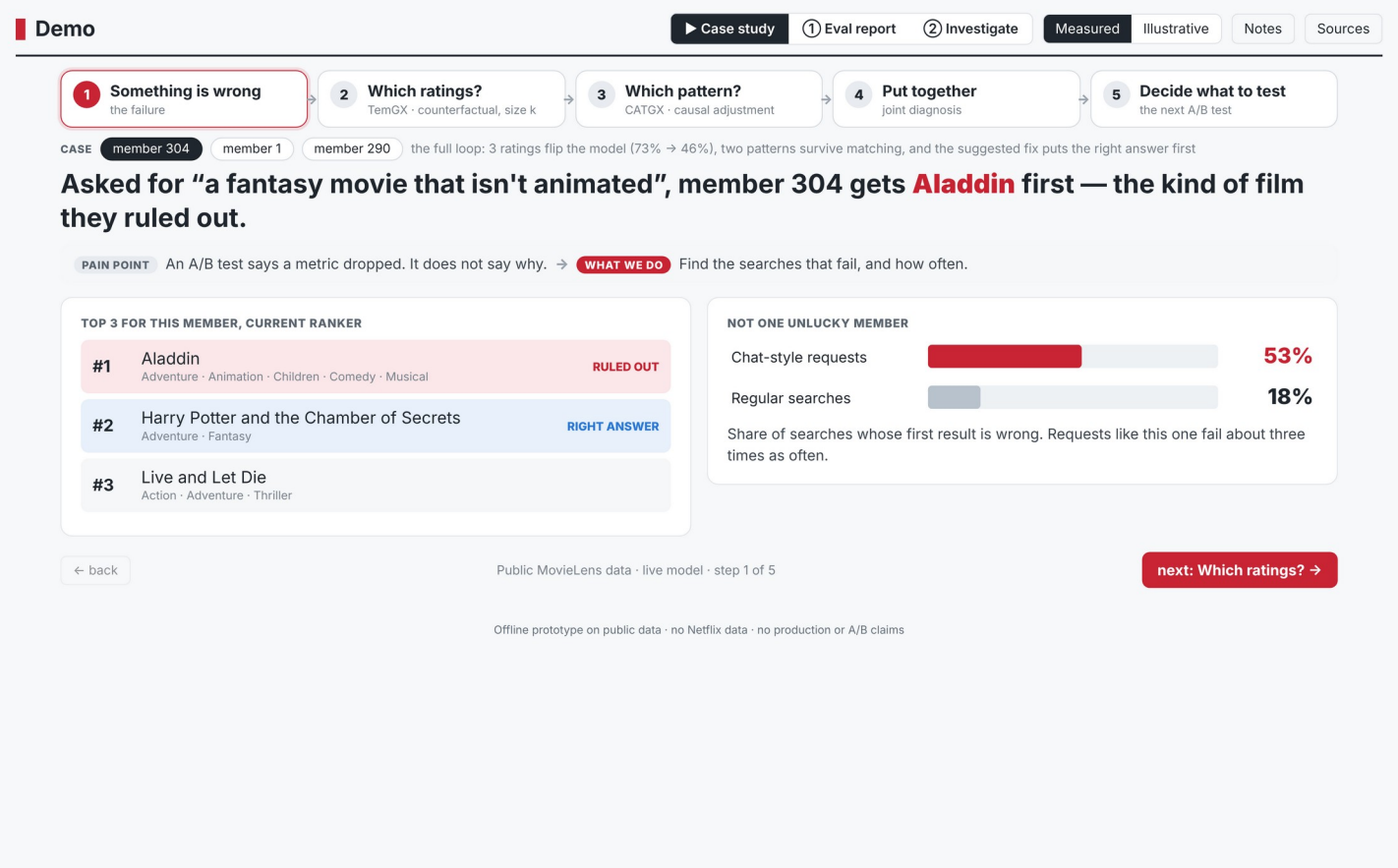


21 probes from 2 planted patterns. Dots = probes. Tick = median fraction of planted effect removed. Exhaustive pairs = upper bound.

What this gives your team: a known reference for testing model-input explanations.

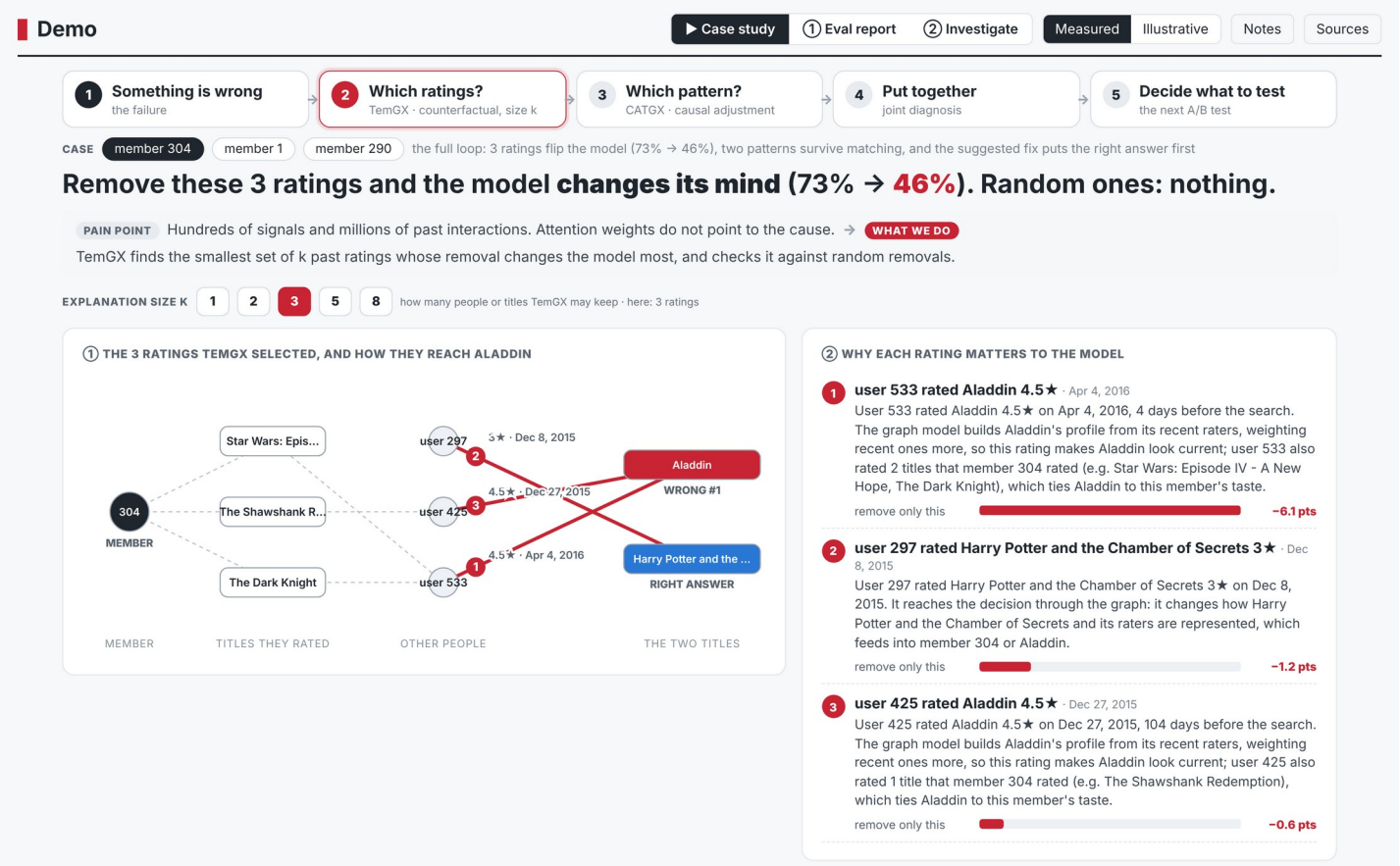
Method: [1]. Experiment: Demo on MovieLens [5]. 21 probes across 2 planted mechanisms; metric is effect-removal fraction.

Demo fallback 1: failed search



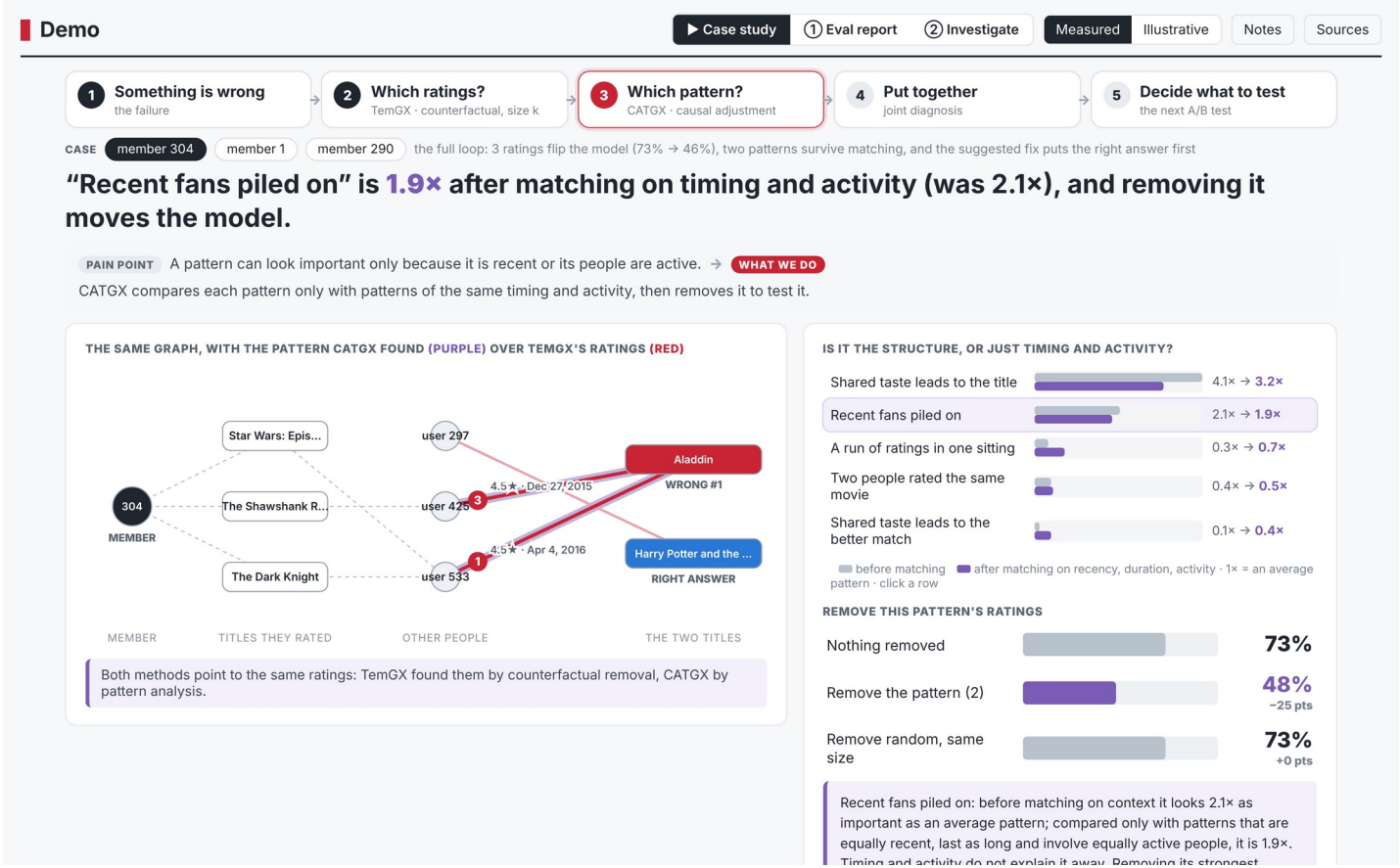
Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 2: historical evidence



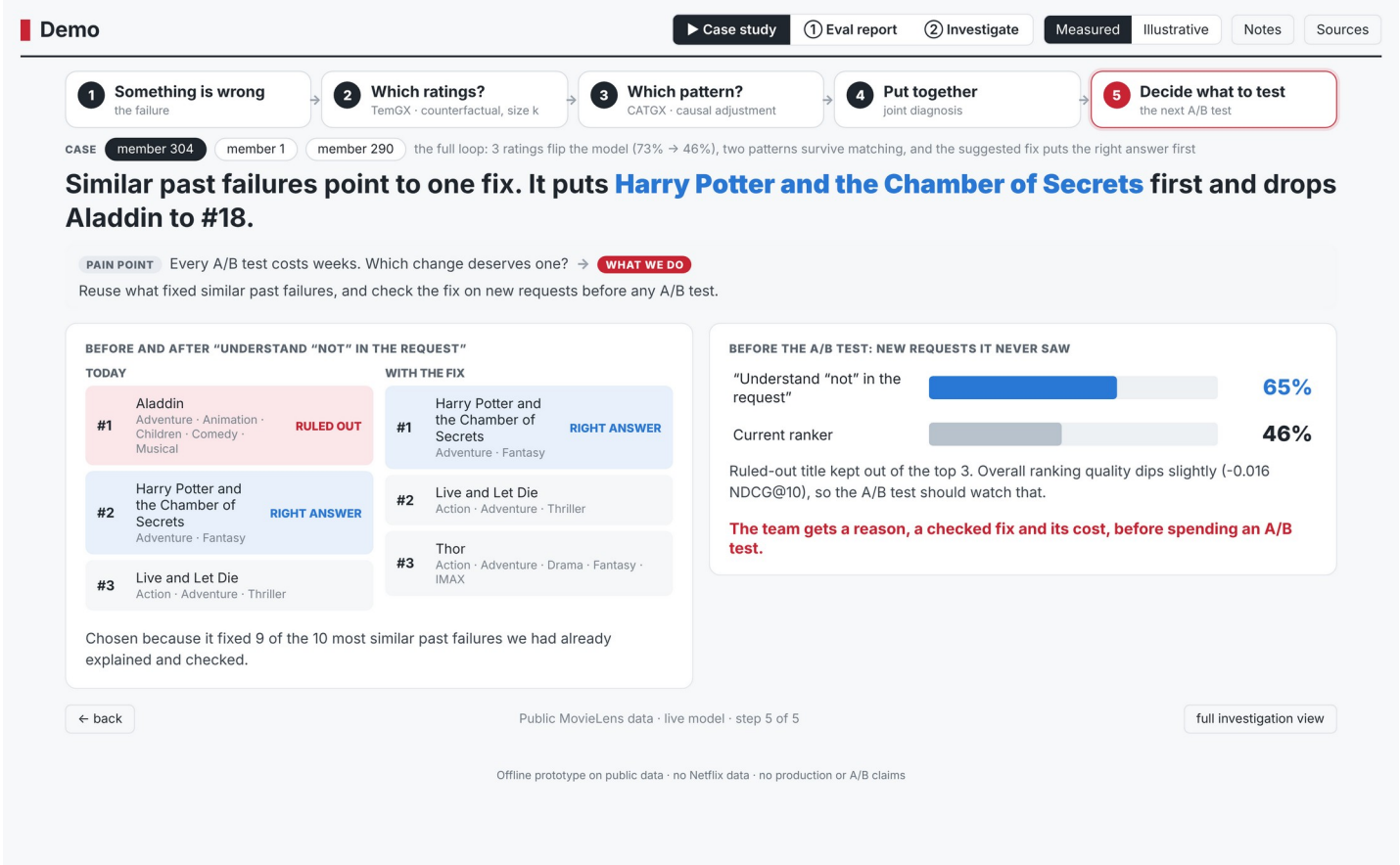
Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 3: deletion check



Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

Demo fallback 4: repair check



Dataset: MovieLens [5]. Demo screenshot, supplied by the author.

References

[1] Lu, M., et al.

Training-free Counterfactual Explanation for Temporal Graph Model Inference. ICLR 2026.

[2] Lu, M., Vu, H., Le, V. K., Ma, J., & Wu, Y.

CATGX: Causal-Aware Temporal Graph Explanation via Scalable Motif Sampling and Adjustment. KDD 2026.

[3] Gomez-Uribe, C. A., & Hunt, N.

The Netflix Recommender System: Algorithms, Business Value, and Innovation. ACM TMIS 6(4), 2015.

[4] Kohavi, R., Longbotham, R., Sommerfield, D., & Henne, R. M.

Controlled experiments on the web: survey and practical guide. Data Mining and Knowledge Discovery 18, 2009.

[5] Harper, F. M., & Konstan, J. A.

The MovieLens Datasets: History and Context. ACM TiiS 5(4), 2015.

[6] Wachter, S., Mittelstadt, B., & Russell, C.

Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR. Harvard Journal of Law & Technology 31(2), 2018.

[7] Pearl, J.

Causal inference in statistics: An overview. Statistics Surveys 3, 2009.

[8] Paranjape, A., Benson, A. R., & Leskovec, J.

Motifs in Temporal Networks. WSDM 2017.

[9] Jain, S., & Wallace, B. C.

Attention is not Explanation. NAACL-HLT 2019.

[10] Wiegreffe, S., & Pinter, Y.

Attention is not not Explanation. EMNLP-IJCNLP 2019.

[11] Malkov, Y. A., & Yashunin, D. A.

Efficient and Robust Approximate Nearest Neighbor Search Using Hierarchical Navigable Small World Graphs. IEEE TPAMI 42(4), 2020.

[12] Lu, M., et al.
Image Processing Pipeline for Fluoroelastomer Crystallite Detection in Atomic Force Microscopy Images. Integrating Materials and Manufacturing Innovation 12, 2023.

All entries are published articles. Demo results are separate author experiments.

References (continued)

[13] Lu, M., et al.
Context Determines Optimal Architecture in Materials Segmentation. ICLR 2026 Workshop on AI for Accelerated Materials Design (AI4Mat). arXiv:2602.04154.

[14] Lewis, P., et al.
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS 2020.

[15] Edge, D., et al.
From Local to Global: A Graph RAG Approach to Query-Focused Summarization. arXiv:2404.16130, 2024.